Asset Allocation Through Retirement in New Zealand: A Simulation and Optimization Approach

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1 Foreword and Research Context

This research is conducted in partnership with Consilium NZ LTD (hereafter, Consilium). Consilium is a New Zealand-based financial services company that provides tools and consulting services to financial advisers and their clients in areas including investment management, portfolio construction, risk management, and financial planning. Consilium also offers managed funds and portfolio solutions designed to support advisers in meeting a wide range of client needs. The aim of this research partnership is to examine and improve lifetime asset allocation strategies for individual investors, with particular focus on retirees in the New Zealand context. Our objective is to produce insights and recommendations that may help Consilium better support advisers and their clients.

Although this research is conducted in partnership with Consilium, the views expressed in this paper are ours and do not necessarily reflect the official policy or position of Consilium. We retain full academic independence in the design, analysis, and reporting of this work. We also retain full ownership of the intellectual property associated with this research, while granting Consilium a non-exclusive licence to use the findings internally and share them with clients and partners. No confidential or proprietary information provided by Consilium has been used in this study.

We thank Consilium for their support and collaboration in this research project, and for the feedback that has helped shape the direction and focus of this work.

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2 Introduction

Lifetime asset allocation is one of the highest-impact financial decisions households make. In general, sources of higher asset returns are associated with higher risks (e.g., (Campbell, 1996; Ghysels et al., 2005; Lundblad, 2007)). This makes the task of selecting an optimal allocation strategy a complex balancing act: weighing the investor's need for growth to fund future spending against their tolerance for volatility and potential losses. The problem is further complicated by the fact that very few of these factors are known in advance. Future returns, future spending needs, and even the investor's own risk tolerance can all evolve over time in response to changing circumstances. This makes the problem of lifetime asset allocation one of making the best possible decisions in the face of deep uncertainty about the future. The answer to this tricky puzzle matters at both household and fiscal levels. The global asset-backed pension market is large (OECD, 2025), and in New Zealand retirement outcomes directly interact with public finances through New Zealand Superannuation. Treasury projections indicate substantial long-run pressure from population ageing (The Treasury, 2025), while policy and industry commentary increasingly emphasises higher private saving and better KiwiSaver decisions (Dexter, 2025; Gilbert, 2025; Luxon, 2025; Palmer, 2025). Better, more efficient, and less risky guidance in this area therefore has private welfare value and public-finance relevance, and this research aims to contribute to both.

Although some investors and advisors spend considerable energy selecting particular securities or funds, Ibbotson and Kaplan (2001) find that roughly 90% of the variation in fund returns is explained by the allocation to major asset classes such as equities, bonds, and cash. As a result, allocation to these broad asset classes is the primary driver of a portfolio's long-term behaviour. This is even more true for typical retirement portfolios (and especially products like KiwiSaver) which are usually held in managed funds, where the investor has little say in the granularity selection of specific securities. KiwiSaver Schemes (with the exception of Consilium's own KiwiWrap and Sharsies as time of writing) typically do not allow the investor to exercise any control over security selection, but instead give them a choice between a small number of pre-defined strategies allocating to different amounts of "Growth" and "Income" asset classes (Financial Markets Authority, 2015). So called "Growth" assets typically comprise a mix of domestic and international equities, while "Income" assets are a mix of domestic and international bonds and cash. In this context, the key decision for most investors is not which specific stocks or bonds to hold, but rather how much of their portfolio to allocate to Growth versus Income assets, and therefore how much risk to take on.

A well performing allocation strategy is one that maximises the investor's expected welfare, which is a function both of the expected return and the protection against adverse outcomes. Where the investor's needs are fixed, Merton (1969) shows that the optimal risky-asset share is constant through time. There are many simplifying assumptions in this result, but probably the most significant for lifecycle investing is that the investor's spending needs are fixed and known in advance. In reality, this is not true as diferential longevity risk, health shocks, and changing preferences all mean that spending needs are uncertain and evolve over time.

Broadly speaking, the investor's lifecycle can be divided into two phases: accumulation and decumulation. During the accumulation phase, the investor is typically working and saving, building up their wealth over time. When they transition to retirement, they enter the decumulation phase, where they draw down on their accumulated wealth to fund their spending needs. This report will focus primarily on the decumulation phase, but the accumulation phase is also relevant for understanding the optimal trajectory of risk over the lifecycle, and so we will spend some time discussing the literature on both phases, especially as it relates to the shape of the optimal glidepath (the trajectory of risk over time).

While Merton (1969) deals specifically with fixed spending needs in decumulation, Modigliani and Brumberg (2005) develops a more general lifecycle model that incorporates both phases and the transition between them. The Life Cycle Hypothesis (LCH) posits that rational households smooth consumption over a finite lifetime by saving during their earning years and drawing down wealth in retirement, with the trajectory of wealth following a characteristic hump shape that peaks around the point of retirement. This model, expanded several times by Modigliani and others (see Modigliani, 1986) provides a enduring theoretical framework for understanding the lifecycle of saving and spending, which is a topic this report will return to in the context of understanding the optimal trajectory of spending and risk over the lifecycle.

With respect to asset allocation across the lifecycle, Bodie et al. (1992) adds in the concept of human capital, which represents the present value of future labor income. When human capital is bond-like, younger households can rationally hold more equity in financial wealth and reduce equity with age (insofar as they continue to be employed). This gives the main theoretical foundation for the intuitive idea that younger investors should hold a higher proportion of their portfolio in risky, high-return assets, and then gradually reduce this proportion as they approach retirement age. This informs the intuitive principles for conventional wisdom around investing over the lifecycle. That is, in essence, that younger investors should take more risk because they have more time to recover from potential losses, while older investors should take less risk to preserve their accumulated wealth as they approach retirement and begin drawing down on their assets. Based in a understanding of these principles, a large number of intuitively appealing, easy-to-communicate age-based rules have emerged that have become the industry standard for retirement investing. These include the various age-in-bonds rules (e.g. the 120-minus-age rule), and target-date funds which are investment products that automatically adjust the asset allocation over time according to a predetermined schedule. Collectively, these types of strategies where asset allocation is a deterministic function of age are often (and hereafter) referred to as "glidepaths". A great deal of work has been committed to studying the optimal shape of these glidepaths, both in terms of their theoretical properties and their empirical performance. They have been developed into a myriad of financial products and are widely recommended by financial advisers and retirement planning tools. A local example is the SuperLife Age Steps strategy, which automatically adjusts the allocation to Growth and Income assets based on the investor's age, with the proportion of Growth assets declining as the investor gets older (see Figure 1).

Although widely recommended, the optimal shape of the glidepath is a subject of active debate in the

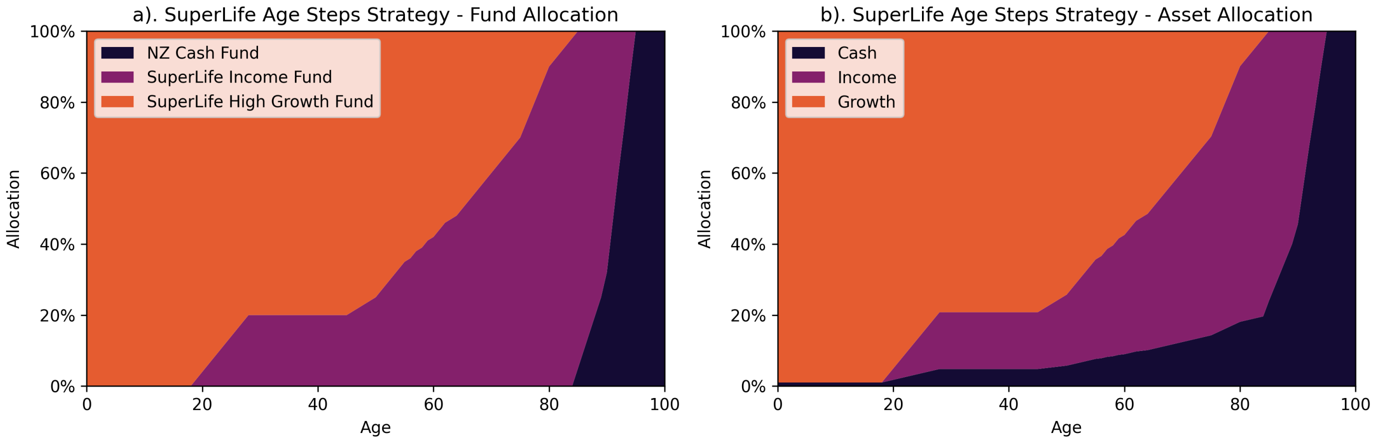

Figure 1: SuperLife's Age Steps target strategy as per their Statement of Investment Policy and Objectives (SIPO), showing the allocation to different funds (a) and the resulting allocation to different asset classes (b). Source: SuperLife (2026).

literature, with different studies reaching different conclusions about whether risk should decline, rise, or remain constant over time, and whether the optimal strategy differs between the accumulation and decumulation phases. The following section will review the literature on this topic, with particular attention to where disagreements arise and how they are driven by modelling choices rather than fundamental empirical disagreement. We will then outline our research questions and approach for addressing these questions in the New Zealand context.

2.1 The Glidepath in Academia

What emerges from the literature is a complex and unresolved debate about the optimal shape of the glidepath, with different studies reaching different conclusions about whether risk should decline, rise, or remain constant over time, and whether the optimal strategy differs between the accumulation and decumulation phases.

There are essentially five main shapes that have been proposed for the optimal glidepath (see Figure **??**):

- Declining equity (DE): the conventional wisdom that risk should decline with age, often implemented as a simple age-based rule.
- Rising equity (RE): the contrarian view that risk should actually increase with age, particularly at the start of retirement, to capture the higher expected returns of equities when they are most needed.
- U-shaped: declining equity during the accumulation phase, followed by rising equity during retirement.
- Inverted U-shaped: rising equity during the accumulation phase, peaking at retirement, followed by declining equity during retirement.
- Static allocation: the view that a constant equity share throughout life is optimal, and that the shape of the glidepath is less important than the overall level of risk taken.

The reasons for this disagreement are complex, but a key fault line in the literature is the choice of methodology for generating return scenarios and evaluating strategies. Different studies use different approaches, including historical rolling periods, Monte Carlo simulation with IID returns, and block bootstrap methods that preserve time-series dynamics. These methodological choices have a significant impact on the results and conclusions drawn about the optimal glidepath shape.

The standard industry recommendation is that investors decrease their risk exposure as they age, continuing to reduce risk even after retirement. This approach appears in investment textbooks (e.g Bengen, 1996; Bodie et al., 2023) and is widely recommended by financial advisers and retirement planning tools (Choi, 2022). A review of U.S target date funds (TDFs) by Dolvin et al. (2010) find that most follow some form of declining-risk structure, often resembling a 120-minus-age equity rule. For a domestic example, take the SuperLife Age Steps allocation profile shown in Figure 1. At one point a declining-risk glidepath was considered as part of potential reforms to the United States Social Security system under the George W. Bush administration (see R. Shiller, 2005; R. J. Shiller, 2006).

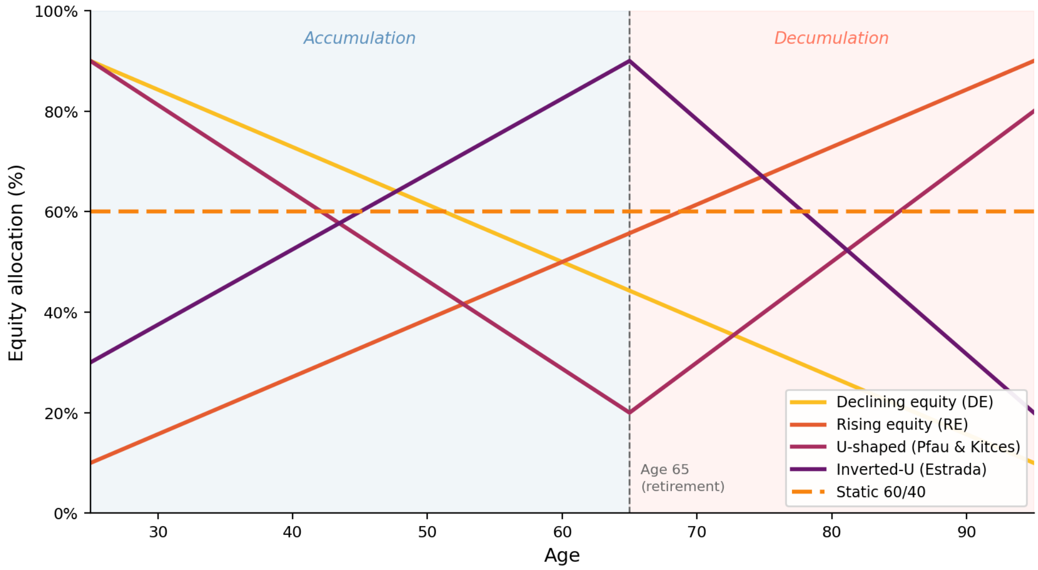

Figure 2: Illustration of different proposed glidepath shapes. The x-axis represents age, with the vertical dashed line indicating the point of retirement. The y-axis represents the equity share of the portfolio. The different lines represent the different proposed glidepath shapes: declining equity (DE), rising equity (RE), U-shaped, inverted U-shaped, and static allocation.

However, declining-risk glidepaths have been challenged by a substantial body of academic work, and the post-retirement period in particular is a site of active and unresolved disagreement in the literature. Responding to the aforementioned Bush-era proposal, R. Shiller (2005) and R. J. Shiller (2006) find that declining-risk glidepaths often fail to outperform a 100% equity strategy. Further work by Estrada (2014, 2015) finds that declining-risk strategies tend to underperform constant-risk alternatives or even contrarian rising-risk strategies across a wide range of countries and historical periods.

On the other hand, studies generating return scenarios using Monte Carlo simulation with IID returns tend to find support for U-shaped glidepaths. This means a declining-risk strategy during the accumulation phase with reduced equity exposure around the point of retirement, and then rising again during the retirement/decumulation phase. The rationale is that retiring with a conservative asset allocation limits the damage from a bad market crash. The proposed risk mechanism that this avoids is a sequence-of-returns argument: the risk that an unlucky a series of bad returns early in retirement depletes the portfolio when it at its largest (in dollar terms). In this line of thinking, you reduce exposure to risk during accumulation, starting at a point of high equity exposure when the investor is young because the sequence risk mechanism is less relevant when the portfolio is still growing and the investor is saving. Proponents of a U-shaped glidepath further argue that the same logic applies in reverse during retirement: starting conservatively at the point of retirement limits the damage from a bad early period, while systematically accumulating equities at depressed prices through a downturn allows the portfolio to capture the eventual recovery when growth is most needed.

Pfau and Kitces (2014) make the case for rising equity during retirement using Monte Carlo simulation. Across 10,000 simulated retirements with lognormal IID returns, 121 glide path combinations (varying start and end equity exposure from 0% to 100%), they find that rising equity during retirement reduces both the probability and magnitude of portfolio failure. The optimal starting allocation is 20–40% equity, rising to 60–80% by end of retirement. In their analysis, a 30–60% rising path beats a static 60/40 on failure rate and failure magnitude. They report that this holds across three capital market assumption sets and argue that retirement outcomes are almost entirely determined by the first 15 years. They argue that a RE strategy during retirement limits the damage from a bad early period and then systematically accumulates equities at depressed prices through the downturn.

Anarkulova et al. (2025) use a block-bootstrap methodology applied to a dataset of monthly real returns across 39 developed countries from 1890 to 2023, drawing on over 31,800 country-months of data and preserving the time-series and cross-sectional structure of historical returns. Under block bootstrap, the optimal allocation is approximately 100% static equity throughout the lifecycle. The sole exception is a brief tactical allocation to bills at the point of retirement, which does resemble a kind of aggressive U-shaped glidepath, but they argue that this is a special case, driven by the rigidity of a fixed real 4% withdrawal rule (based on Bengen, 1994) and when spending flexibility is restored, this exception disappears and the all-equity result holds at every age. They also argue that the conventional preference for bonds is substantially overstated by return-generation decisions in the literature. They compare optimal glidepaths under IID return sampling and block bootstrap, and find that the optimal bond allocation is higher under IID assumptions. They argue that this is because the long-horizon properties of bonds are much less favourable than their short-horizon statistics suggest. At 30-year horizons, bond variance *increases* relative to the one-year level (variance ratio 2.30), whereas equity variance *decreases* (variance ratio 0.75). Bonds also carry a correlation of -0.78 against inflation, which they argue destroys real purchasing power in precisely the environments most harmful to retirees. In essence, Anarkulova et al. (2025) argue that IID assumptions overstate the risks of sequence-of-return risk. In reality, timeseries effects mean the sequence is actually autocorrelated and mean-reverting. In this environment, the higher expected returns of equities are more reliably captured over long horizons and the risks of bonds are more pronounced.

Adding more complexity to the debate, Duarte et al. (2021) use a deep reinforcement learning (DRL) approach to solve a comprehensive lifecycle model that incorporates both accumulation and decumulation phases, with spending needs that evolve over time in response to shocks to health, longevity, and preferences. They use Monte Carlo simulated returns with autocorrelated timeseries dynamics, and find that the optimal equity allocation at retirement is approximately 60%, and that TDFs (which land at 30–40% equity) are too conservative; the welfare cost of following a TDF rather than the optimal policy is 1.7–2.8% of lifetime consumption. However, they otherwise find that a declining trajectory of equity allocation accross the lifecycle is broadly optimal, and that the optimal trajectory post-retirement is approximately flat (not declining or rising).

And while it may appear that the DE structure is broadly supported during the accumulation phase, work by Estrada (2014) using historical rolling periods across 19 countries finds that DE strategies have higher failure rates than contrarian RE mirrors in every specification. This means that in their analysis of the accumulation phase, a more optimal strategy is to start with a lower equity share when the portfolio is small and then increase it as the portfolio grows, which is the opposite of the conventional wisdom. They argue that a DE strategy lowers the expected return of the portfolio precisely when the portfolio is at its largest: lifecycle funds are aggressive when the portfolio is small (early career) and conservative when it is large (near retirement), which is backwards. The companion paper, Estrada (2015), applies the same dataset and methods to the retirement drawdown period, testing 81 rolling 30-year retirement windows starting from 1900 with a 4% real withdrawal rule. The key finding reverses the accumulation result: DE strategies have lower failure rates *and* higher terminal wealth than rising-equity mirrors in every specification. In the United States, DE failure rates range from 4.9 to 8.6%, while the corresponding RE mirrors fail at rates of 8.6 to 21.0%. Across all 19 countries, DE fails at 31–32% versus RE at 39–49%. Combined with their accumulation findings, Estrada's recommendation for a full-lifecycle glidepath shape is an *inverted U*: equity should increase during working life, peak at the point of retirement, and then decline through the drawdown period.

At this point, it is clear that the literature is deeply divided on the optimal glidepath shape, and that these disagreements are driven by methodological choices rather than fundamental empirical disagreement. The choice of return-generation methodology (historical rolling periods, Monte Carlo with IID returns, block bootstrap) appear to have a significant impact on the results and conclusions drawn about the optimal glidepath shape. The choice of risk measure (failure probability, expected shortfall, risk-adjusted success ratio, expected utility) and the specification of the spending rule (fixed real withdrawal, flexible spending, etc.) also matter. However, the takeaway from this literature is that there is almost no consensus on the optimal glidepath shape. This calls into question the value of the glidepath concept itself, and whether dynamic allocation strategies that adjust risk exposure soley on the basis of age or time in retirement are actually optimal and can effectively manage the hetrogeneous asset return paths that the cohorts of investors in these studies face. In reality, the glidepath debate may be a red herring, and the choice of overall risk level (i.e. the equity share) may be more important than the shape of the trajectory. Blanchett (2007) tests 43 glide paths (11 static allocations and 4 types of declining-equity dynamics (linear, stair-step, concave, and convex)) bootstrapped from 80 years of monthly return data, across 1,071 scenarios defined by distribution period (20–40 years) and real withdrawal rate (3–8%). Rising-equity paths are not tested; the comparison is strictly static versus DE. Static allocations have the lowest failure probability in all but one of the 1,071 scenarios. The best static portfolio by raw failure probability is 100% equity; using his "Success to Variability Ratio" (probability of success divided by portfolio standard deviation), the best risk-adjusted portfolio is a 60/40 stocks-bonds allocation.

An important observation from Blanchett (2007) is that the choice of glidepath matters little when withdrawal rates are moderate and horizons are short. At a 4% withdrawal rate over a 20-year period, the range of failure probabilities across all 43 strategies is only 1.56 percentage points. The same metric reaches 62.45 percentage points at a 6% withdrawal rate over 40 years. According to their findings, the glidepath debate is therefore most consequential for aggressive scenarios, and the literature's disagreements should be interpreted in that light. Anarkulova et al. (2025) and Estrada (2014, 2015) also make the case for a 100% equity static allocation, arguing that the risks of sequence-of-returns risk are overstated in the literature and that the long-run properties of equities are more favourable than bonds.

The tension between these results is not easily resolved, but it is clear that the choice of return-generation methodology is a key driver of disagreement. Historical rolling periods (Estrada, Bengen) preserve the actual structure of return sequences. Proponents of this technique argue that because recessions cluster, valuations mean-revert, and bad periods tend to be followed by recoveries, IID assumptions overstate the risks of sequence-of-returns risk as stocks are more likely to recover from an early dip than the IID model suggests. This approach does, however, limit the sample size to the number of non-overlapping historical periods, which is a particular problem for long-horizon retirement planning where the number of 30- or 40-year periods in the historical record is very small. Anarkulova et al. (2025) use a block bootstrap approach, in which blocks of returns of a random length (with an average of 12 months) sampled from the historical record and concatenated to create synthetic return paths. This preserves some of the time-series dynamics of returns, including mean reversion and volatility clustering, while generating a much larger sample of data. However, even in this approach and with the exceptionally long historical record afforded to them through the GFD dataset, the number of unique return paths is still limited by the historical record and compounded by data quality issues, particularly for the early part of the record. A result of this may be that equity returns become overstated due to survivorship bias (companies that went bankrupt or were delisted may be missing from the data). Duarte et al. (2021) use a Monte Carlo approach with autocorrelated return dynamics, which might arguably be more realistic than IID assumptions, but still relies on a parametric model of return dynamics that may not capture all relevant features of the data. We also have methodological concerns with using a neural network to create asset allocation policies in response to the same state variables used to generate the return paths, which may lead to over-fitting and type of model-data leakage that overstates the performance of the resulting policies. We also find it difficult to interpret the resulting policies and understand the mechanisms driving their performance, which limits the insights that can be drawn from this approach.

The choice of objective function or performance metric also could influence the diagreement. Pfau and Kitces (2014) specifically target failure risk minimisation; Estrada (2014, 2015) targets both failure risk and terminal wealth jointly and finds DE dominates on both. Blanchett (2007) introduces the risk-adjusted Success to Variability ratio and finds it tilts the optimal allocation toward moderate equity, regardless of glidepath shape. Studies optimising expected utility with explicit risk aversion parameters (e.g. Duarte et al. (2021)) tend to find approximately constant, high equity allocations consistent with neither the U-shape nor the inverted-U, but broadly aligned with Blanchett's static recommendation.

Finally, the specification of the investor's spending and income needs is a critical driver of optimal policy. In discussing the paper by Anarkulova et al. (2025), Felix (2025) argues that the real risk to investors is not the risk of portfolio failure per se, but the risk of being unable to adjust spending in response to market conditions.

2.2 Consilium's Position

Consilium does not have an official position on the optimal glidepath shape nor does it currently offer any specific guidance or products that implement a particular glidepath strategy. Consilium's approach to retirement planning is focused on providing tools and insights that help financial advisers and their clients make informed decisions about asset allocation and withdrawal strategies during retirement. However, an adviser using Consilium's official products and recommendations could construct a glidepath strategy themselves by combining the recommendations from two different areas of Consilium's offering: the Model Portfolios and the Goal-Positioning Software (GPS). For this reason, much of the analysis in this report will be contextualised with reference to these two areas of Consilium's offerings.

"Model Portfolios" are predesigned asset allocation strategies for different risk profiles following Consilium's proprietary asset allocation methodology, which is based primarily on the Fama-French three-factor model (Fama & French, 1993). There are 42 model portfolios across five families of strategies: Classic, SRI, Synergy Classic, Synergy SRI, and Synergy PIE. These portfolios are designed to cater to a range of investor preferences and risk tolerances, with varying allocations to Growth and Income assets. A glidepath approach could be constructed from these portfolios by tatically switching between them at different ages, but currently there is no official recommendation or guidance on how to do this, and the portfolios themselves are not explicitly designed to be used in this way. The availability of portfolios across risk levels by family is shown in Table 1.

The Goal-Positioning Software (GPS) is a tool that helps advisers and their clients set and track progress toward financial goals, including retirement. It provides personalised projections of future wealth and income based on the client's current and expected contributions, asset allocation (to one of the model portfolios), and

Table 1: Availability of portfolio models across risk levels by family

Risk Level (Growth/Income)	**Synergy PIE**	**Synergy Classic**	**Synergy SRI**	**SRI**	**Classic**
20/80	–	x	x	x	x
30/70	x	x	x	x	x
40/60	x	x	x	x	x
50/50	x	x	x	x	x
60/40	x	x	x	x	x
70/30	–	x	x	x	x
80/20	x	x	x	x	x
90/10	–	x	x	x	x
98/02	x	x	x	x	x

other factors. The GPS allows users to simulate different scenarios and see how changes in their savings rate, asset allocation, or other assumptions might affect their ability to meet their retirement goals. Central to the GPS is a Monte Carlo simulation engine that generates a large number of potential future return paths based on the historical performance of the model portfolios, and uses these simulations to estimate the probability of achieving the client's retirement goals under different scenarios. The GPS does not currently have any specific features or recommendations related to glidepath strategies, but it does allow the user to change their allocation at any point in time, which could be used to implement a glidepath strategy if the user chooses to do so.

The GPS is beset by the same modelling choices that drive disagreement in the literature, and so it is important to understand how these choices affect the recommendations it produces. For example, the GPS uses a Monte Carlo simulation engine with IID asset returns generated through Brownian Motion, which may lead to different conclusions about optimal glidepath shape compared to a block bootstrap approach that preserves time-series dynamics. Additionally, the performance of the GPS is lacking, and it is currently unable to compute more than around 1,000 simulated return paths in a reasonable time frame, which limits the robustness of its recommendations. This is an area where improvements could be made and while it is not the focus of this research, we will note any areas of our approach that could be implemented in the GPS to improve its performance and robustness.

2.3 Problem Statement and Research Questions

The review above reveals that the literature has not converged on a single answer to even the most basic question of optimal retirement allocation, and that disagreements are substantially driven by modelling choices rather than fundamental differences in the underlying economic problem. The choice of return-generation methodology (IID Monte Carlo, historical rolling periods, or block bootstrap), the choice of risk or objective measure (failure probability, terminal wealth, expected utility), and the specification of the spending rule (fixed real withdrawal vs. flexible spending) each independently shift the apparent optimal policy. A secondary but practically important question is whether the glidepath shape matters at all compared with the more fundamental question of whether bonds should be held in retirement portfolios at the levels currently recommended.

Concretely, we wish to answer the following questions:

1. Are conventional age-based declining-risk glidepaths optimal under standard assumptions? And does that result survive under NZ-calibrated return dynamics and spending paths that decline in real terms with age?

2. How sensitive are "optimal" recommendations to modelling assumptions (return-generation process, IID vs. block bootstrap, risk measure, spending rule)?

3. What state time-varying policies (e.g., wealth-responsive) outperform fixed asset allocation strategies, and how do they perform under different modelling assumptions?

4. What, if any, robust policy recommendations can be made for the New Zealand context, given the institutional setting of NZ Superannuation and the growing role of KiwiSaver?

2.3.1 Our Approach and Specification

We make three contributions relevant to the New Zealand context and Consilium's position on this topic. First, we review the New Zealand-specific literature on retirement and retirement policy, and outline the needs and behaviour of New Zealand retirees. Second, we develop a method for generating lifetime wealth and spending paths for a representative New Zealand retiree, calibrated to the institutional context of New Zealand Superannuation and KiwiSaver. We aim to improve on the current model used by Consilium in the GPS and other areas. Third, we will develop an optimistation framework to create dynamic asset allocation strategies and compare their performance under different modelling assumptions.

3 Retirement in New Zealand

The New Zealand-specific literature on retirement concerns remains limited, so we draw extensively on international evidence. Understanding the institutional context is essential for interpreting this evidence appropriately. New Zealand's retirement system combines a universal public pension (New Zealand Superannuation) with voluntary private savings through KiwiSaver. New Zealand Superannuation provides all residents aged 65 and over with a baseline income regardless of work history or assets, and adjusts annually for inflation. KiwiSaver allows voluntary contributions from employees, matched by employers and the government. Together, these mechanisms provide a safety net while encouraging personal saving (Ministry of Social Development, 2022; Te Ara Ahunga Ora: Retirement Commission, 2021).

KiwiSaver was introduced in 2007, so most current retirees have little or no accumulated balance. Currently, 40 percent of New Zealanders aged 65 rely entirely on New Zealand Superannuation, with a further 20 percent holding minimal additional savings (Retirement Commission Te Ara Ahunga Ora, 2022). As the scheme matures and government policy encourages higher contribution rates (Luxon, 2025), future cohorts will retire with substantially larger balances. This demographic shift increases the need for robust guidance on asset allocation and withdrawal strategies during retirement.

Given these trends, the Australian experience becomes increasingly relevant. Australia's retirement system combines a means-tested public pension with mandatory superannuation contributions of 12 percent of income (Australian Taxation Office, n.d.; Swoboda, 2014). While institutional differences remain, the growing role of KiwiSaver makes the two systems more comparable than in the past.

3.1 New Zealand Superannuation Policy Settings

New Zealand Superannuation (NZ Super) is a universal, non-means-tested public pension paid from age 65, funded from general taxation rather than a contributory insurance scheme. It forms the guaranteed income floor underpinning all retirement planning in New Zealand. Entitlement is not conditional on prior employment, investment, or KiwiSaver participation; a recipient's private wealth or income has no effect on the payment rate.

Eligibility

The primary eligibility criteria are age and residency. A person becomes entitled upon turning 65, provided they have been both resident and present in New Zealand for a sufficient period since age 20. For those born on or before 30 June 1959, the requirement is 10 years total, of which 5 must have accrued since age 50. For those born on or after 1 July 1977, the total requirement is 20 years (with 5 since age 50), reflecting a graduated increase introduced by the Social Assistance (Residency Qualification) Legislation Act 2018. Special provisions apply to recognised refugees and to periods spent in the Cook Islands, Niue, and Tokelau (New Zealand Parliament, 2001, s. 8).

Payment Rates

Rates are set in Schedule 1 of the New Zealand Superannuation and Retirement Income Act 2001 and adjusted each 1 April by Order in Council. The April 2026 rates (before deduction of tax under code M) are listed in Table 2.

After deduction of standard tax (code M), the fortnightly after-tax rates are approximately \$1,110 for a single person living alone and \$1,708 combined for a couple, equivalent to roughly \$28,900 and \$44,400 per year respectively (Work and Income New Zealand, 2026a, 2026b). These figures differ slightly from those used

Table 2: NZ Super rates from 1 April 2026 (before tax). Source: Work and Income New Zealand (2026a, 2026b).

Rate type	Per week	Per fortnight	Per year
Single living alone	$647.37	$1,294.74	$33,663
Single sharing	$595.57	$1,191.14	$30,970
Couple (per person)	$492.14	$984.28	$25,591
Couple (combined)	$984.28	$1,968.56	$51,182

in Matthews (2025) (who report $519 and $799 per week for singles and couples respectively) due to differing tax rate assumptions; this study uses the WINZ statutory rates as the primary reference.

Indexation and the Wage Floor

The Act mandates annual adjustment for CPI inflation (so the rate cannot decline in real terms) subject to a binding wage floor. The combined after-tax couple rate must remain between 66 and 72.5 percent of net average ordinary-time weekly earnings, as measured by the Quarterly Employment Survey (New Zealand Parliament, 2001, s. 16). In practice, wages have grown faster than CPI in recent decades, meaning the wage floor has been the binding constraint. The single living alone rate is fixed by statute at 65 percent of the couple combined rate (after tax), and the single sharing rate at 60 percent. This legislated floor is a key feature distinguishing NZ Super from defined-benefit pension schemes that index only to prices.

Policy Adjustments and Reform Debate

Despite ongoing fiscal pressure from population ageing, NZ Super's core parameters have remained largely unchanged since 2001. The most significant recent amendment was the gradual increase in the residency qualifying period from 10 to 20 years for those born after 1 July 1977, phased in from 2018. The eligibility age of 65 has not changed; a proposal to raise it to 67 was advanced and subsequently abandoned. A 2022 government review (the Retirement Review Issues Paper, conducted by Te Ara Ahunga Ora) examined a range of possible adjustments including age, means-testing, and residency, without proposing binding changes (Ministry of Social Development, 2022).

Supplementary Benefits

Several additional transfers are available to eligible retirees:

- **Winter Energy Payment:** a universal supplement paid automatically to NZ Super recipients from 1 May to 1 October each year. The 2026 rate is $20.46 per week for single people and $31.82 per week for couples and people with dependants (Work and Income New Zealand, 2026a). No application or means test is required.
- **Accommodation Supplement:** a means- and asset-tested housing cost subsidy available to renters, boarders, and mortgage holders across all age groups including retirees. The amount varies by accommodation costs and location zone, with higher rates in Auckland and Wellington. Eligible retirees with moderate housing costs may receive up to several hundred dollars per fortnight (Work and Income New Zealand, 2026a).
- **SuperGold Card:** a concession card issued automatically to NZ Super recipients, providing discounts on public transport (including free off-peak travel on most regional services), selected retail, and other services (Ministry of Social Development, 2022).
- **Residential Care Subsidy:** an asset-tested subsidy for long-term residential aged care (rest homes and dementia units). As at 2022, a single person is generally required to contribute their own assets up to a threshold of around $240,000 before the subsidy takes effect, with the family home exempt for a defined period if a spouse, dependent, or carer remains in residence (Ministry of Social Development, 2022).

Together, NZ Super plus the Winter Energy Payment provide a reliable inflation-linked income floor for all retirees.

3.2 Patterns of Spending in Retirees - New Zealand and Abroad

3.2.1 Spending By Age

The Stats NZ Household Economic Survey (HES) provides the most comprehensive data on New Zealand retirement spending patterns. Two recent reports are particularly relevant. Le and Richardson (2023) analyse expenditure patterns by age across New Zealand households, while Retirement Income Interest Group (2024) focus specifically on spending through retirement and its implications for drawdown strategies.

Using HES data for the year ended June 2019, Le and Richardson (2023) find that expenditure declines sharply at retirement and continues to fall thereafter. Mean household expenditure is $64,700 for ages 65–74, then 28% lower for ages 75–84, and 44% lower for ages 85+. Expenditure declines across most categories, with the largest falls in mortgage repayments, air travel, and recreation. Unusually, healthcare expenditure also declines with age in New Zealand, likely reflecting the public funding of most core medical services. Retirement Income Interest Group (2024) estimate the real decline at roughly 2% per year after age 65.

The HES data cannot, however, distinguish voluntary from involuntary spending reductions. Retirees may reduce expenditure because they become less active and have genuinely lower needs, or because they face binding financial constraints. Without longitudinal data tracking individuals over time, these explanations cannot be separated cleanly. The observed decline likely reflects both channels.

Understanding the source of this decline is central to evaluating welfare. If spending falls primarily due to financial constraints, optimal strategies should aim to sustain pre-retirement consumption levels. If it reflects voluntary lifestyle adjustment, forcing consumption to remain high may be neither necessary nor welfare-improving. This distinction also affects risk tolerance: financially constrained retirees may need safer portfolios, while those adjusting voluntarily may tolerate more risk.

As we have mentioned earlier, beggining with work by Modigliani and Brumberg (2005), and expanded several times by Ando and Modigliani (1963), and others, the Life-Cycle Hypothesis (LCH) establishes a theoretical or "ideal" pattern of saving and consumption over the life course. Under the LCH, individuals are expected to smooth consumption over time, in other words, they wish to essentially maintain a stable standard of living throughout their life. The LCH predicts that individuals will save during their working years, accumulating wealth to fund consumption in retirement (but only saving insofar as it does not unnecessarily reduce their pre-retirement consumption) and then draw down that wealth during retirement to maintain their living standards, ultimately exhausting their wealth near the end of life. The widely observed decline in spending at retirement (and lower-than-expected wealth drawdown), therefore, is puzzling under the LCH framework, and has been termed the "retirement consumption puzzle" by various authors. Understanding the source of this decline has been the topic of numerous papers and many explainations have been proposed.

Proponents of the LCH might argue that the observed decline in spending at retirement is primarily driven by financial constraints, which compel retirees to reduce their consumption below their desired level. This could be due to insufficient savings, unexpected health shocks, or other factors that limit retirees' ability to maintain their pre-retirement standard of living. However, the widely observed tendancy of retirees to not only reduce spending but also to not draw down on their wealth suggests that the decline in spending may not be solely due to financial constraints. King (1982) in a survey of Canadian retirees finds that the majority of respondents did not only fail to draw down on their wealth, but many actually accumulated more wealth during retirement. mThis finding has been corroborated by a number of subsequent studies in many different countries and time-periods (e.g. De Nardi et al., 2010; Love et al., 2009; Palumbo, 1999). A paper by the Grattan Institute in Australia (Grattan Institute, 2019) finds that fewer than half of Australian retirees draw down wealth at all, and over 40% accumulate wealth during retirement. Moreover, it finds that most Australian retirees are not financially constrained: retirees report feeling more financially secure than working-age households.

So whether how much of the observed decline in spending at retirement is driven by financial constraints versus voluntary lifestyle adjustment is an open question, but it does not appear that financial constraints are the primary driver. Consumption expenditure is an imperfect proxy for welfare, particularly when higher spending is driven by medical needs or lower spending reflects voluntary lifestyle adjustment. While Barrett and Brzozowski (2010) find that households retiring unexpectedly (and are thus unprepared) due to adverse health shocks do experience sharper spending declines than those retiring as planned (which is consistent with involuntary constraints driving some of the observed decline), the majority of the observed decline may be voluntary. Retirees spend less on commuting, work clothing, and purchased meals (Aguiar & Hurst, 2008; Battistin et al., 2009), and may substitute market purchases with home production, reducing measured expenditure while maintaining consumption (Becker, 1965; Hurd & Rohwedder, 2003). These mechanisms can mask substantial cross-household heterogeneity in retirement outcomes. Consistent with this interpretation is that the largest expenditure declines observed in the HES data by Le and Richardson (2023) are in discretionary categories such as transport and recreation, while spending on essentials such as food and housing remains comparatively more stable.

Cross-country comparisons highlight the role of institutional context. Banks et al. (2016) find that U.K. retirees reduce real consumption by 2.2% per year versus 1.4% per year in the U.S. This divergence is driven largely by healthcare financing: U.S. retirees face high out-of-pocket costs that rise with age, while U.K. retirees benefit from comprehensive public coverage. When healthcare spending is excluded, consumption trajectories are more similar across countries.

Australian evidence also shows spending declines of 0.5–1% per year in real terms (Grattan Institute, 2019). Unlike New Zealand's cross-sectional HES data, the Grattan Institute analysis uses longitudinal data tracking individuals over time. The decline persists in both cross-sectional and longitudinal samples, suggesting it reflects genuine behavioural change rather than cohort effects. As Retirement Income Interest Group (2024) note, the weight of evidence indicates that spending reductions primarily reflect voluntary adjustment to changing lifestyle needs rather than binding constraints. From our perspective, this means that it is reasonable to model retirement spending as declining in real terms with age.

3.2.2 Bequests and Precautionary Savings

An aspect of the "retirement consumption puzzle" that is particularly troubling for the LCH is that a large number of retirees die with substantial wealth intact. This situation is often described by financial planners as a kind of risk, that is that a retiree may outlive their wealth and have thus died "underconsuming" relative to their potential. A common explanation for this phenomenon is that retirees intentionally preserve wealth to leave bequests to their heirs (e.g. Hurd, 1987). In this view, retirees are not underconsuming but are instead optimally consuming while intentionally the "utility" of leaving a bequest to their heirs. However, the empirical evidence for intentional bequest motives is weak.Alonso-García et al. (2022) find that Australian and Dutch retirees have little to no intentional bequest motive outside of provision for a surviving spouse, and that the observed pattern of non-decumulation is more consistent with saving for uncertain future costs, such as aged care or health shocks. This motive is often referred to as a "precautionary" motive, and is distinct from a bequest motive in that the utility of leaving a bequest is not derived from the act of leaving a bequest itself, but rather from the desire to avoid the risk of being unable to cover future costs.

Precautionary motives are well-supported empirically. De Nardi et al. (2010) show that exposure to uncertain health and long-term care costs explains much of the observed non-decumulation in the United States. Similarly, Banks et al. (2016) find that healthcare costs drive higher late-life spending among U.S. retirees relative to those in the U.K. with broader public coverage. New Zealand's system is mixed: most core medical services are publicly funded, but long-term care such as assisted living and memory care can involve substantial out-of-pocket costs (New Zealand Government, n.d.). Retirement Income Interest Group (2024) conclude that precautionary motives in New Zealand are likely weaker than in the United States but stronger than in countries such as France and Germany with more comprehensive public provision, placing New Zealand alongside the United Kingdom in a mixed-funding model. However, aged care is a significant cost to

3.2.3 Spending by Household Composition and Wealth

Household composition and housing tenure are among the strongest predictors of retirement spending levels and subjective wellbeing in New Zealand. Le and Richardson (2023) find that single-person retired households spend around $30,700 per year on average, compared with $65,100 for couples. Notably, spending increases more than proportionately with household size: the average couple spends more than double the amount of a single-person household, implying weaker economies of scale (in New Zealand at least) than other studies, such as (Anarkulova et al., 2025), have suggested in other contexts. Le and Richardson (2023) attribute the disproportionate gap primarily to differences in discretionary spending: couples devote substantially more to transport and recreation and culture, the two largest discretionary categories. Single-person households living alone are also more likely to be renters or to carry a mortgage, further constraining their discretionary budget, while couples are more likely to own outright.

Homeownership status has a particularly pronounced effect on both spending and wellbeing. Among retired New Zealand households, approximately 66 percent own their home outright, 14 percent carry a mortgage, and 14 percent rent (Le & Richardson, 2023). Outright owners report the highest subjective wellbeing of any tenure group and spend around $55,800 per year, of which housing costs form a relatively small share. Renters spend considerably less in total, at around $41,400 per year, but pay approximately $13,100 of this in rent annually, leaving a significantly lower discretionary budget. Renters also report the lowest wellbeing among all tenure types, even after controlling for income. This suggests that the financial and psychological burden of ongoing housing costs in retirement is substantial and is not fully compensated by lower total spending. This well-being gap will have interesting consequences for asset allocation decisions, especially as it pertains to whole-life allocation and spending strategies. One of the peculiarities of the New Zealand KiwiSaver system is that it allows for the use of accumulated savings to be withdrawn for the purchase of a first home. Balancing the competing goals of homeownership and retirement savings is a unique challenge for New Zealand investors, and one that may have important implications for their lifetime asset allocation decisions. For instance, if utility gained from homeownership is sufficiently high, it may justify depleting the investors retirement savings while young to achieve homeownership, even if this results in a lower savings balance at retirement. Inversely, if buying a house depletes the investor's retirement savings to a degree that the excess burden of housing costs would have otherwise have been more than offset by the additional savings balance at retirement, it may be optimal to delay homeownership and instead prioritise retirement savings.

The importance of homeownership is expected to increase over time. Retirement Income Interest Group (2024) note that home ownership among New Zealanders aged 65 and over is projected to fall from around 80 percent today to approximately 60 percent by 2048, as younger cohorts face higher barriers to entry in the housing market. This shift implies that expenditure benchmarks calibrated on the current cohort of retirees (who benefit from unusually high ownership rates) will understate spending requirements for future generations, particularly with respect to housing costs.

3.2.4 Expenditure Benchmarks for New Zealand Retirees

The most widely used empirical benchmarks for New Zealand retirement spending are the annual Retirement Expenditure Guidelines published by the NZ Fin-Ed Centre at Massey University (Matthews, 2025). These are derived from the Statistics New Zealand Household Economic Survey (HES) and reported at two tiers: a "No Frills" budget based on the second income quintile of retired HES households, reflecting a basic standard of living with few luxuries, and a "Choices" budget based on the fourth quintile, reflecting a more comfortable lifestyle with some discretionary spending. The 2025 figures, adjusted to 30 June 2025, are shown in Table 3.

Table 3: New Zealand Retirement Expenditure Guidelines, 2025. Weekly spending, with after-tax NZ Super rates for comparison. Source: Matthews (2025).

	One-person household		Two-person household	
Budget	Metro	Provincial	Metro	Provincial
No Frills ($/week)	705	581	937	1,061
Choices ($/week)	791	772	1,780	1,243
NZ Super after tax ($/week)	519 (single)		799 (couple)	

For the No Frills provincial single-person household, NZ Super alone covers approximately 89 percent of spending, implying a required annual drawdown of only around $3,200 from private savings. For the No Frills metro single, the gap is somewhat larger at around $9,700 per year. Couples at the Choices tier, particularly in metro areas, face substantially higher gaps; the Choices metro couple budget of approximately $92,600 per year implies a drawdown requirement of around $51,000 annually above NZ Super. However, this upper estimate should be interpreted cautiously, since the fourth income quintile of HES retired households likely includes a material proportion of households still receiving employment or investment income, rather than pure retirees drawing down accumulated savings.

A central limitation of these guidelines is that they are constructed from cross-sectional data and therefore implicitly treat retirement spending as fixed in real terms throughout retirement. The HES samples households of all retirement ages at a single point in time, which means the guidelines effectively average the higher spending of the more numerous newly retired with the lower spending of older cohorts, without capturing the within-person spending trajectory over time (Retirement Income Interest Group, 2024). If real spending declines by approximately 2 percent per year after age 65 (as both Retirement Income Interest Group (2024) and Le and Richardson (2023) find) then the guidelines will overstate the savings balance required at age 65 for any given income target.

Retirement Income Interest Group (2024) quantify this directly. Using a replacement-rate methodology, they calculate that reaching a total income target of $56,000 per year in real terms for a single retiree, on top of NZ Super, over a retirement from age 65 to 90, requires a savings balance of \$605,000 under a constant-real-spending assumption. If instead nominal income is assumed to remain flat (equivalent to a 2 percent real decline when inflation is also 2 percent), the required balance falls to \$375,000, a reduction of 40 percent. An intermediate assumption of a 1 percent real annual decline yields a balance of \$480,000. These calculations assume a 3.5 percent net investment return, which is not particularly high and is broadly consistent with the long-run real return on a relatively conservative portfolio.

For the purposes of this study, we treat the expenditure guidelines as an empirical anchor for age-65 spending levels. We calibrate initial retirement spending to the No Frills and Choices benchmarks as the lower and upper bounds for a representative retiree, and model real consumption as declining at approximately 2 percent per year thereafter. NZ Super is treated as a guaranteed, inflation-linked income floor from age 65. Under this calibration, the required drawdown from financial assets diminishes progressively as retirement spending falls toward the level already covered by NZ Super. This pattern has material implications for both the required savings balance at retirement and the optimal asset allocation strategy during decumulation.

4 Methods for Simulating and Optimising Lifetime Wealth Trajectories

The crux of our analysis is the ability to simulate realistic wealth trajectories for individual investors. This in and of itself is not a novel contribution, but the specific features of our simulation framework are designed to accommodate the particular questions we seek to answer. The goal, however, is to develop a flexible, modular framework that can be extended beyond the specific questions of this thesis to support ongoing research and policy analysis for Consilium and others. In this respect, while the specific questions we ask in this thesis are narrowly focused on the optimality of age-based glidepaths in the New Zealand retirement context, the framework we develop is intended to be more broadly applicable to a wide range of questions in the lifetime asset allocation space, including questions about optimal whole-life allocation and spending strategies, the role of housing and other non-financial assets, and the interaction between retirement savings and other life goals such as homeownership. To this end, much of this section will describe the general structure of the simulation framework, with specific details about the calibration to the New Zealand context and the particular questions we ask in this thesis reserved for later sections. Included also, in the appendix of this thesis will be possible extensions to the framework that could be implemented in future work, such as the inclusion of housing and other non-financial assets, and the modelling of homeownership decisions and their interaction with retirement savings and spending.

The problem of simulating lifetime wealth trajectories has essentially two components: the methods used to simulate investor behavoiur and the methods used to simulate return dynamics. The former includes the specification of the asset allocation strategy, the spending rule, and the risk measure used to evaluate outcomes. The latter includes the choice of return-generation methodology (IID Monte Carlo, historical rolling periods, or block bootstrap). In this section, we will describe the general structure of our simulation framework, and how it accommodates different choices in these areas. We will also discuss the advantages and disadvantages of different approaches, and how we have calibrated our framework to the New Zealand context.

4.1 Lifecycle Simulation Framework

With the exception of the continuous-time approach pioneered by Merton (1969), the majority of the literature on simulating lifetime wealth trajectories uses discrete-time approaches that model the investor's wealth at regular intervals, such as annually or monthly (e.g. Anarkulova et al., 2025; Estrada, 2014; Mäkinen & Toivanen, 2024). This approach is far easier to implement and makes less assumptions about the underlying return process than the continuous-time approach, which relies on strong assumptions about the return process (for example, that returns are normally distributed and independent over time) that are not supported by empirical evidence.

In this framework, we evolve the investor's wealth over time according to a simple recursive transition equation that accounts for returns, contributions, and withdrawals. For the transition from time t to $t+1$, the investor's wealth is updated as follows in (equation 1):

$$w_{t+1} = g_{t+1}w_t + \pi_{t+1} \tag{1}$$

Where w_t is the wealth at time t, g_{t+1} is the growth factor on the portfolio between time t and $t+1$, and π_{t+1} is the net contribution (or withdrawal if negative) made at time $t+1$. This can represent regular contributions during the accumulation phase or withdrawals to cover spending needs during the decumulation phase.

The growth factor g_{t+1} is determined by the asset allocation strategy being tested, which specifies the proportion of wealth invested in different asset classes. Assuming that the portfolio is rebalanced at the end of each period, the growth factor can be calculated as follows in equation 2:

$$g_{t+1} = 1 + \mathbf{a}_t^\top \mathbf{r}_{t+1} \tag{2}$$

Where $\mathbf{a}_t$ is the asset allocation vector at time t, with each element $a_{i,t}$ representing the proportion of wealth invested in asset class i, and $\mathbf{r}_{t+1}$ is a vector of returns for each asset class between time t and $t+1$. Both $\mathbf{a}_t$ and $\mathbf{r}_{t+1}$ are vectors of length k, where k is the number of asset classes in the portfolio.

In a Monte Carlo simulation, we generate a large number of possible investor wealth trajectories by simulating returns across multiple paths. The returns are stored in a tensor $\mathcal{R}$ of dimension $N \times T \times k$, where N is the number of simulated paths, T is the number of time periods, and k is the number of asset classes. Similarly, the allocation policy can be represented as a tensor $\mathcal{A}$ of dimension $N \times T \times k$, where each element $A_{i,t,j}$ represents the allocation to asset class j at time t for simulation path i. In our nomenclature, asset allocation policies are functions or sets of rules that determine the allocation vector $\mathbf{a}_t$ at each time t. By representing the allocation policy as a tensor, we can accommodate a wide range of policies, including those that depend on path-specific state variables such as wealth or age. For example, a simple age-based glidepath can be represented by setting all paths to have identical allocations at each time, while a more complex policy that adjusts allocations based on the investor's current wealth can be represented by allowing the allocations to vary across paths at each time step.

At each time $t+1$, we extract the $N \times k$ matrix $\mathbf{R}_{:,t+1}$ containing returns at time $t+1$ across all paths and asset classes, and the $N \times k$ matrix $\mathbf{A}_{:,t}$ containing allocations at time t for all paths. The growth factors for all paths are then computed in vector form as shown in equation 3:

$$\mathbf{g}_{t+1} = \mathbf{1}_N + \sum_{j=1}^{k} \mathbf{R}_{:,t+1,j} \odot \mathbf{A}_{:,t,j} \tag{3}$$

Where $\mathbf{1}_N$ is an $N \times 1$ vector of ones, $\mathbf{R}_{:,t+1,j}$ and $\mathbf{A}_{:,t,j}$ are the $N \times 1$ column vectors for asset class j, $\odot$ denotes element-wise multiplication, and $\mathbf{g}_{t+1}$ is the resulting $N \times 1$ vector of growth factors across all paths. Equivalently, this can be expressed as $\mathbf{g}_{t+1} = \mathbf{1}_N + \text{rowsum}(\mathbf{R}_{:,t+1} \odot \mathbf{A}_{:,t})$, where the element-wise product is summed across the asset class dimension. The wealth evolution across all paths then follows equation 4:

$$\mathbf{w}_{t+1} = \mathbf{w}_t \odot \mathbf{g}_{t+1} + \boldsymbol{\pi}_{t+1} \tag{4}$$

Where $\mathbf{w}_t$, $\mathbf{g}_{t+1}$, and $\boldsymbol{\pi}_{t+1}$ are vectors of length N representing the wealth, portfolio returns, and contributions or withdrawals for each of the N simulated paths at time t, and $\odot$ denotes element-wise (Hadamard) multiplication. The wealth across all paths over all time periods can be represented as an $N \times T$ matrix, $\mathbf{W}$, where each row corresponds to a single simulated path of wealth over time.

Note that the system of equations above is agnostic to the return-generation process. The returns tensor $\mathcal{R}$ is a flexible input that can be generated using any methodology. This allows us to test the robustness of our findings across different assumptions about return dynamics, such as using historical returns, bootstrapped returns, or model-simulated returns. Additionally, $\mathbf{a}_t$ and $\boldsymbol{\pi}_{t+1}$ can be specified as functions of time, wealth, or other state variables, allowing for a wide range of asset allocation and spending strategies to be simulated within the same framework. This flexibility is a key strength of our approach, as it allows us to explore a broad set of strategies and assumptions without needing to fundamentally alter the underlying simulation mechanics. It is on the construction of these functions and the calibration of the return-generation process that we will focus in the next sections.

Implications for Consilium: Matrix Operations and Computational Efficiency

The effort taken to express the model in matrix form is not only for mathematical clarity; it also allows us to leverage vectorised operations through highly optimised linear algebra libraries (for example, `numpy` (Harris et al., 2020)) to vectorise the the calculations across the matrices longest dimension (the path dimension). This means that we can compute the wealth evolution across all paths virtually simultaneously using efficient matrix operations, rather than iterating through each path sequentially with for-loops. This vectorised approach can take advantage of modern CPU architectures and optimised libraries to perform computations much faster than a non-vectorised implementation, which can substantially accelerate simulation when path counts and horizon lengths are large. Compared to the GPS, which takes upwards to 30 seconds to simulate less than 1000 paths over a 30-year horizon, our matrix-based implementation can simulate 100,000 paths over a 50-year horizon in under a second on a standard laptop. This dramatic improvement in computational efficiency enables us here to explore a much wider range of strategies with much more vigour than would be possible with the GPS tool, and also opens the door to more complex strategies that may require more computational resources, such as those that involve path-dependent allocations or more sophisticated risk measures.

Refactoring the GPS tool to use matrix operations would be a significant improvement in its computational efficiency and flexibility. It would allow Consilium to support more complex strategies and larger simulations, which could significantly enhance the tool's value for financial advisors and their clients.

4.2 Modelling Contributions and Withdrawals

The flow of wealth into and out of the portfolio is critical not only for the evolution of system, but also for the evaluation of outcomes. During retirement, when our investors is dissaving to fund consumption, how much the investor is withdrawing and is able to withdraw is a key determinant of welfare. Modelling contributions and withdrawals in a realistic way is therefore essential for simulating realistic wealth trajectories and evaluating the performance of different asset allocation strategies. There have been many approaches to modelling and planning withdrawals in the literature and financial advice space.

Our framework consists of essentially three components: the withdrawal rule, which specifies how much the investor desires to withdraw in each period; the consumption floor, which specifies the minimum amount the investor must withdraw to meet essential needs; and the income, which specifies the guaranteed income from sources such as NZ Super. The withdrawal rule is the most commonly discussed component in the literature, and there are many different rules that have been proposed. However, on its own it is not sufficient to model the withdrawal behaviour of a representative retiree, as it does not account for the fact that retirees have minimum spending needs that must be met regardless of market returns, and that these needs are not captured by any of the simple rules. The income is also important, as it provides a guaranteed source of income that can help to meet essential needs and reduce the risk of depleting wealth too quickly.

Net inflows, π_t, are modelled by equation 5 as the difference between the income, y_t, and realised consumption, $\hat{c}_t$.

$$\pi_t = y_t - \hat{c}_t \tag{5}$$

Note that we will constrain $\pi_{t+1} \geq -W_t$ such that the investor cannot withdraw more than their current wealth balance (go into debt). This is a realistic constraint, as it is not possible for a retiree to withdraw more than their current wealth balance. In practice, retirees may have access to credit or other sources of funds that could allow them to temporarily withdraw more than their current wealth balance, but this is not a sustainable strategy and would likely lead to financial distress in the long run. By imposing this constraint, we ensure that our simulations reflect the realistic limitations faced by retirees in managing their withdrawals.

4.2.1 Inflation

Central to any model of consumption is the treatment of inflation. In the context of retirement planning, inflation is a critical factor that erodes the purchasing power of money over time and therefore has a significant impact on the real value of withdrawals and the sustainability of retirement savings. Inflated dollar amounts can also be a significant source of confusion, as retirees and planners may have different perceptions of what constitutes a reasonable withdrawal amount depending on whether they are thinking in "real" (inflation-adjusted) or "nominal" (actual dollar) terms. Real consumption refers to the purchasing power of the retiree's spending, while nominal consumption refers to the actual dollar amount spent. The two are related by the inflation rate, which erodes the purchasing power of money over time.

In our framework, we denote the rate of inflation at time t as ι_t, and the cumulative inflation factor from time 0 to time t as $s_t = \prod_{i=1}^{t}(1 + \iota_i)$. Real dollar amounts can be converted to nominal dollars by multiplying some initial real amount by the cumulative inflation factor. At $t = 0$ (the point of retirement), real and nominal dollar amounts are the same, but as time progresses, the real value of a given nominal amount will decline if there is positive inflation. For example, if a retiree desires to maintain a constant real consumption level of \$50,000 per year, and cumulative inflation over the first three years of retirement is 10%, then the nominal withdrawal amounts in year three would need to be \$55,000 to maintain the same real purchasing power.

While there are technically many different types of inflation which will affect different parts of a retiree's overall consumption bundle differently, for the purposes of this analysis we will treat inflation as a single aggregate factor (linked to the consumer price index) that affects the overall purchasing power of the retiree's withdrawals. This is a simplification, but it is a common approach in the literature and is sufficient for our purposes here.

Implications for Consilium: Inflation

Another major weakness of the GPS tool is that it only accounts for inflation in a very limited way. The tool allows the user to specify an assumed inflation rate, but this is a constant rate that is applied uniformly across all time periods and does not account for the stochastic nature of inflation or its potential correlation with asset returns. This means that the GPS tool cannot capture the impact of unexpected inflation shocks on the real value of withdrawals or the sustainability of retirement savings, which is a critical aspect of retirement planning.

There are very good reasons to be wary of this approach. As we have seen over the 2020-2024 period, inflation can be highly volatile. Additionally, as noted by Anarkulova et al. (2025), inflation can be correlated with asset returns, particularly due to the fact that inflation is often addressed by central banks increasing interest rates, which can have a negative impact on asset returns. Monetary policy has a complex relationship with both fixed income and equity returns. For example, rising interest rates can lead to lower bond prices and higher yields, which can reduce the value of fixed income investments. Rising interest rates can also lead to lower equity prices, as they increase the cost of borrowing for companies and reduce the present value of future cash flows. By not accounting for the stochastic nature of inflation and its potential correlation with asset returns, the GPS tool may provide an overly optimistic view of the sustainability of retirement savings and the real value of withdrawals, particularly in periods of high or volatile inflation.

4.2.2 Modelling Withdrawal Rules

Withdrawal rules are probably the most widely discussed component of retirement planning in the financial advice space, and there are many different rules that have been proposed. Perhaps the most widely known is the "4% rule" popularised by Bengen (1994), which suggests that a retiree can safely withdraw 4% of their initial retirement savings balance in the first year of retirement, and then adjust that amount for inflation in subsequent years, without running out of money over a 30-year retirement period.

In their report, Retirement Income Interest Group (2024) propose several alternative withdrawal rules tailored to the New Zealand context, each reflecting different assumptions about spending trajectories through retirement. We briefly describe these rules here in terms of the desired consumption at time t, denoted c_t^*. Each of these rules is designed to be simple enough for retirees to implement without sophisticated planning tools, and each implicitly or explicitly responds to the empirical observation that real spending declines through retirement. Retirement Income Interest Group (2024) note that the 6% rule is consistent with the observed 2% annual real decline when general inflation is also 2%, as it holds nominal spending approximately constant (after the asset returns are accounted for).

The 6% Rule:

This rule specifies a fixed nominal withdrawal amount equal to 6% of the initial retirement balance:

$$c_t^* = 0.06 \times w_0 \tag{6}$$

This rule front-loads spending and is well-aligned with the empirical finding that real spending declines by approximately 2% per year after age 65. When general inflation is also 2%, holding nominal spending constant produces the observed real decline. However, because withdrawals are fixed in nominal terms while returns are stochastic, there is a risk that the fund will be depleted before death.

The Inflated 4% Rule:

This is a variant of Bengen's original formulation, where the initial 4% withdrawal is adjusted annually for inflation:

$$c_t^* = 0.04 \times w_0 \times s_t \tag{7}$$

where s_t is the cumulative inflation factor at time t since retirement. This rule maintains constant real spending, which is inconsistent with the empirical evidence of declining consumption but is more conservative than the

6% rule. It is more likely to preserve wealth through retirement and may be appropriate for retirees with lower risk tolerance or longer expected lifespans.

The Fixed Date Rule:

This rule divides the current wealth balance by the number of years remaining until a specified terminal age T, at which point the retiree is assumed to rely solely on NZ Super:

$$c_t^* = \frac{w_t}{T - t} \tag{8}$$

This rule produces withdrawals that vary from year to year as wealth fluctuates with market returns. It maximises spending over the chosen horizon but does not account for longevity risk beyond the terminal age. The approach is simple to implement but may produce significant year-to-year volatility in consumption.

The Life Expectancy Rule:

This rule divides current wealth by the actuarial life expectancy at the retiree's current age t:

$$c_t^* = \frac{w_t}{e_t} \tag{9}$$

where e_t is the remaining life expectancy at age t. This approach is dynamically efficient in the sense that, in expectation, it exhausts the fund exactly at death. Because life expectancy declines with age, the denominator shrinks over time, which tends to stabilise the withdrawal rate even as wealth declines. This is perhaps the most economically rational of the simple heuristics, as it balances current consumption against future needs in a way that accounts for mortality risk.

4.2.3 Consumption Floors and Minimum Desired Consumption

While the withdrawal rules described above provide a useful starting point for modelling retirement spending, they do not account for the fact that retirees have minimum spending needs that must be met regardless of market returns. For example, if a retiree has a minimum real spending requirement of $30,000 per year to cover essential expenses, then the 6% rule would fail to meet this requirement if the initial retirement balance is less than $500,000. Similarly, the Fixed Date Rule and Life Expectancy Rule could produce withdrawals that are too low to meet essential needs in years when wealth is depleted by poor market returns.

To model this more realistically, we specify a minimum real spending requirement or floor, $\underline{c}_t$, that must be met in each period. This gives us the minimum desired consumption at time t, denoted c_t, as the maximum of the desired consumption from the withdrawal rule, c_t^*, and the consumption floor, $\underline{c}_t$, as shown in equation 10:

$$c_t = \max\left(c_t^*, \underline{c}_t\right) \tag{10}$$

This consumption floor is designed such that it can be variable over time, or even with respect to other state variables such as wealth or health status. For example, the floor could be set to reflect the empirical expenditure benchmarks for New Zealand retirees, which vary by household composition and housing tenure. The floor could also be adjusted for inflation or other factors that affect the cost of living over time. Random shocks to the floor could also be introduced to reflect unexpected expenses, such as health shocks or changes in housing costs. Based on emprical evidence, Retirement Income Interest Group (2024) suggest that the spending guidelines from Matthews (2025) could decline in real terms by approximately 2% per year after age 65. This can be modelled as in equation 11 as follows:

$$\underline{c}_t = \underline{c}_0 \times (1 - 0.02)^t \times s_t \tag{11}$$

Where $\underline{c}_0$ is the initial real consumption floor at age 65, and s_t is the cumulative inflation factor at time t since retirement.

4.2.4 Income

While certain types of assets pay dividends or interest that can be considered as income, we treat these as part of the portfolio returns rather than as a separate income stream. Income within our framework refers to exogenous sources of income that are not derived from the portfolio. Within a New Zealand retirement context, the most important source of exogenous income is NZ Super, which provides a guaranteed, inflation-linked income floor from age 65. We denote income at time t as y_t.

NZ Super provides a guaranteed, inflation-linked income floor from age 65. The income recieve from NZ Super is calculated within our framework as follows in equation 12:

$$y_t = y_0 \times s_t \tag{12}$$

Where y_0 is the initial NZ Super payment at for the household at age 65 (see table 2), and s_t is the cumulative inflation factor at time t since retirement.

Extensions to the Framework: Modelling Other Sources of Income and the Whole Life Cycle

While our current framework focuses on retirement and the decumulation phase, it can be easily extended to model the whole life cycle, including the accumulation phase and other sources of income. For example, we could model labour income during the accumulation phase, which would allow us to simulate the entire life cycle of wealth accumulation and decumulation. We could also model other sources of income during retirement, such as part-time work, rental income from property, or annuity payments. This would provide a more comprehensive view of the retiree's financial situation and allow us to evaluate how different asset allocation strategies interact with these various sources of income throughout the life cycle.

In a whole-life cycle model, a good model of labour income is critical, as it is the primary source of wealth accumulation for most individuals and one that is also highly heterogeneous across individuals and simulation paths. Modelling labour income, and particularly shocks to labour income such as unemployment or disability, would allow us to capture the risks that individuals face during the accumulation phase and how these risks interact with their retirement savings and asset allocation decisions. This income, and the shocks to it, would also have correlations with other economic variables, such as asset returns and inflation. This model is well outside the scope of the work presented here, but it is a natural extension of the framework we have developed and would be a valuable area for future research.

4.2.5 Realised Consumption

Because our retirees are forbidden from withdrawing more than their current wealth balance, it is possible that the desired minimum consumption, c_t, exceeds the available resources. In such cases, the actual consumption is limited by the wealth balance and any exogenous income, such as NZ Super. The realised consumption at time t, denoted $\hat{c}_t$, is therefore given by the minimum of the desired consumption and the total available resources, as shown in equation 13:

$$\hat{c}_t = \min\left(c_t, w_t + y_t\right) \tag{13}$$

Net contributions to the portfolio are then given by:

$$\pi_t = -\hat{c}_t + y_t \tag{14}$$

This realised consumption is one of the most important outputs of our simulation, as it reflects the actual spending that the retiree is able to achieve given their wealth trajectory and the withdrawal rules. It is also a critical input for evaluating the performance of different asset allocation strategies, as it directly affects the retiree's welfare and the sustainability of their retirement savings. By modelling realised consumption in this way, we can capture the trade-offs that retirees face in balancing current consumption against future needs, and we can also evaluate how different asset allocation strategies perform under different withdrawal rules and income scenarios.

Implications for Consilium: Modelling Consumption and Income

The GPS currently only outputs the investor's wealth trajectory over time, and does not model or output the actual consumption that the retiree is able to achieve given their wealth trajectory and any exogenous income. This is a significant limitation, as it means that the GPS cannot directly evaluate the welfare implications of different asset allocation strategies, since welfare is ultimately determined by the retiree's consumption rather than their wealth.

This weakness is in part due to the limited and inflexible way that the GPS models withdrawals, which does not allow for the kind of dynamic withdrawal strategies described above, and also does not account for the fact that retirees have minimum spending needs that must be met regardless of market returns. While it has a fairly faithful implementation of the NZ Super rules, it does not integrate this income into the overall model of consumption and wealth evolution in a way that allows for the kind of dynamic interactions described above.

4.3 Asset Allocation Policies

The asset allocation policy is the key decision variable in our analysis, and the one that we seek to optimise. Here, a "policy" refers to a function or set of rules that determine the allocation vector $\mathbf{a}_t$ at each time t. The policy can be as simple as a fixed allocation that does not change over time, or it can be a complex, dynamic strategy that adjusts allocations based on the investor's current wealth, age, or other state variables. We express the policy as a function that maps from the current state of the system (for example, time, wealth, or other state variables) and a set of policy settings to the asset allocation vector $\mathbf{a}_t$. We denote this policy function as $\mathcal{A}$, which takes as input the current state and "policy settings", denoted θ, and outputs the allocation vector $\mathbf{a}_t$ for each time t and simulation path. The specific form of the policy function will depend on the particular class of policies we are testing, but the key point is that the goal of finding an optimal asset allocation strategy can be framed as an optimisation problem over the policy parameters θ, where we seek to find the set of parameters that maximises our chosen risk measure of the resulting wealth trajectories.

Our approach borrows heavily from the control-matrix approach of Mäkinen and Toivanen (2024), which provides a flexible and interpretable way to represent a wide range of asset allocation policies, including those that depend on path-specific state variables such as wealth or age. Another notable example of a flexible policy representation is the use of neural networks, as proposed by Duarte et al. (2021), which can capture complex, non-linear relationships between state variables and asset allocations. However, while neural networks offer a high degree of flexibility, the policy settings they produce are not easily interpretable. Because a optimal solution is likely to be linked to the return-generation process we use, we want to be able to understand the structure of the optimal policy and how it responds to different state variables. The control-matrix approach allows us to do this, as the policy settings are directly interpretable as allocations for different combinations of the various state variables fed into the it.

In the control-matrix approach, the policy settings θ take the form of a matrix or tensor where each dimension corresponds to a different state variable that the policy can depend on. It is worth spending some time discussing how we see the control-matrix approach fitting into the broader landscape of asset allocation policy representations in the literature, as our interpretation of this approach will also inform how we implement and interpret the results of our optimisation. The dimesions of the control matrix correspond to different state variables that the policy can depend on. In Mäkinen and Toivanen (2024), the control matrix has two dimensions: time and wealth. This allows for policies that can adjust allocations based on both the investor's current age (or time since retirement) and their current wealth level. This is an extension of the more traditional age-based glidepath, which only allows for dependence on time, which is in turn an extension of a fixed allocation strategy, which does not allow for any dependence on state variables.

To show you what we mean by this, consider the following example. A fixed allocation strategy can be represented as a policy function that does not depend on any state variables, and simply returns the same allocation vector $\mathbf{a}$ at each time t and for each simulation path. In this world, the policy function can be represented as a constant function:

$$\mathcal{A}(\theta) = \mathbf{a} \quad \text{for all } t \text{ and all paths} \tag{15}$$

In a two-asset portfolio, θ would simply be the allocation to the risky asset, with the allocation to the safe asset determined residually, e.g. $\mathcal{A}(\theta) = (a, 1-a)$. For a portfolio with N asset classes, θ would be a vector of length $N-1$ representing the allocations to $N-1$ of the asset classes, with the allocation to the remaining asset class determined residually as one minus the sum of the other allocations. We denote this tranformation as

$$\varphi(\boldsymbol{\theta}) = \boldsymbol{a} = \begin{pmatrix} \theta_1 \\ \theta_2 \\ \vdots \\ \theta_{N-1} \\ 1 - \sum_{i=1}^{N-1} \theta_i \end{pmatrix} \tag{16}$$

Note that we constrain our asset allocation policies to only allow for long positions in the available asset classes, which is a realistic assumption for KiwiSaver investors, as they are not allowed to short-sell or use leverage. This means that the allocation vector $\mathbf{a}_t$ must satisfy the following constraints:

This means that the allocation vector $\mathbf{a}_t$ must satisfy the following constraints:

$$\sum_{j=1}^{k} a_{j,t} = 1 \quad \text{and} \quad a_{j,t} \geq 0 \quad \text{for all } j \tag{17}$$

By extension, our policy settings θ must satisfy the following constraints to ensure that the resulting allocations are valid:

$$\sum_{i=1}^{N-1} \theta_i \leq 1 \quad \text{and} \quad \theta_i \geq 0 \quad \text{for all } i \tag{18}$$

Traditional glidepath strategies vary asset allocations with age (or equivalently, time since retirement). In this case, the policy function can be represented as a function of time:

$$\mathcal{A}(\theta, t) = \boldsymbol{a}_t \tag{19}$$

Where $\boldsymbol{a}_t$ is the allocation vector at time t, which is determined by the policy settings θ. In a very simple case, θ could be a $T \times (k-1)$ matrix where each row corresponds to a time period and each column corresponds to an asset class (except for the safe asset, which is determined residually). This is essentially the form of the policy function used in Anarkulova et al. (2025), where discrete asset allocations are specified for each time period, and the optimal policy is found by searching over the space of possible allocations at each time period. In this case, $\mathcal{A}$ becomes:

$$\mathbf{a}_t = \mathcal{A}(\theta, t) = \varphi(\theta_t) \tag{20}$$

A key insight from Mäkinen and Toivanen (2024) is that you can reduce the parameter space of the policy function (and by extension, extend the policy function to continuous state variables) by using interpolation to determine the asset allocation for time periods that fall between the defined grid points. For example, if we have discrete allocations defined at time periods 0, 5, and 10, we can use linear interpolation to determine the allocation at time period 3 by interpolating between the allocations at time periods 0 and 5. This allows us to have a more parsimonious parameterisation of the policy function, while still allowing for a high degree of flexibility in the shape of the asset allocation strategy over time.

Formally, suppose we have policy settings θ defined at a discrete set of time nodes $\{t_0, t_1, \ldots, t_M\}$, where θ is now a matrix of dimension $M \times (k-1)$ with each row θ_i specifying the risky asset allocations at node t_i. For any time t such that $t_i \leq t < t_{i+1}$, the interpolated allocation is given by:

$$Z\mathbf{a}_t = \mathcal{A}(\theta, t) = \varphi\left((1-\alpha)\theta_i + \alpha\theta_{i+1}\right) \tag{21}$$

where the interpolation weight α is computed as:

$$\alpha = \frac{t - t_i}{t_{i+1} - t_i} \tag{22}$$

This linear interpolation ensures that the policy varies smoothly between node points while requiring only $M \times (k-1)$ parameters rather than $T \times (k-1)$ if allocations were specified independently at every time period. The transformation φ is then applied to the interpolated risky allocations to obtain the full allocation vector including the safe asset.

Our policy function specifies the allocation to each asset class at each time t for each simulation path, and is denoted by the operator $\mathcal{A}$. New Zealand investors cannot leverage or short-sell, so we restrict our attention to policies that only allow for long positions in the available asset classes. Because short-selling and leverage are not allowed, we can reduce the dimensionality of the problem by treating one of the asset classes as the "safe" asset, with the allocation to this asset class determined residually as one minus the sum of the allocations to the other asset classes. For example, if we have two asset classes (a risky asset and a safe asset), we can represent the allocation to the risky asset as a_t and the allocation to the safe asset as $1 - a_t$. This reduces the number of decision variables in our optimisation problem and simplifies the search for optimal policies.

This is our interpretation of a glidepath strategy, which is a policy that specifies the asset allocation as a function of time. A linear glidepath can be represented by θ with two nodes (or rows), one at the point of retirement ($t = 0$) and one at the terminal age ($t = T$), with the allocations at these nodes representing the initial and final allocations, respectively. More complex glidepaths can be represented by adding more nodes and allowing for non-linear interpolation between them. Adding more nodes allows for more complex shapes. By varying the number of nodes, we can also determine the sensitivity of the optimal policy to the granularity of the time dimension, which can provide insights into how important it is to have a finely-tuned glidepath versus a more simple one.

Figure 3 illustrates this interpolation process for a three-asset portfolio (stocks, bonds, and cash) with three time nodes at $t = 0, 15, 30$. Panel (a) shows the policy settings matrix θ, where each row specifies the allocations to stocks and bonds at a discrete time node, with the cash allocation computed residually via φ. Panel (b) shows the resulting continuous allocation surface obtained by linear interpolation, with the vertical dashed lines marking the discrete nodes where θ is defined.

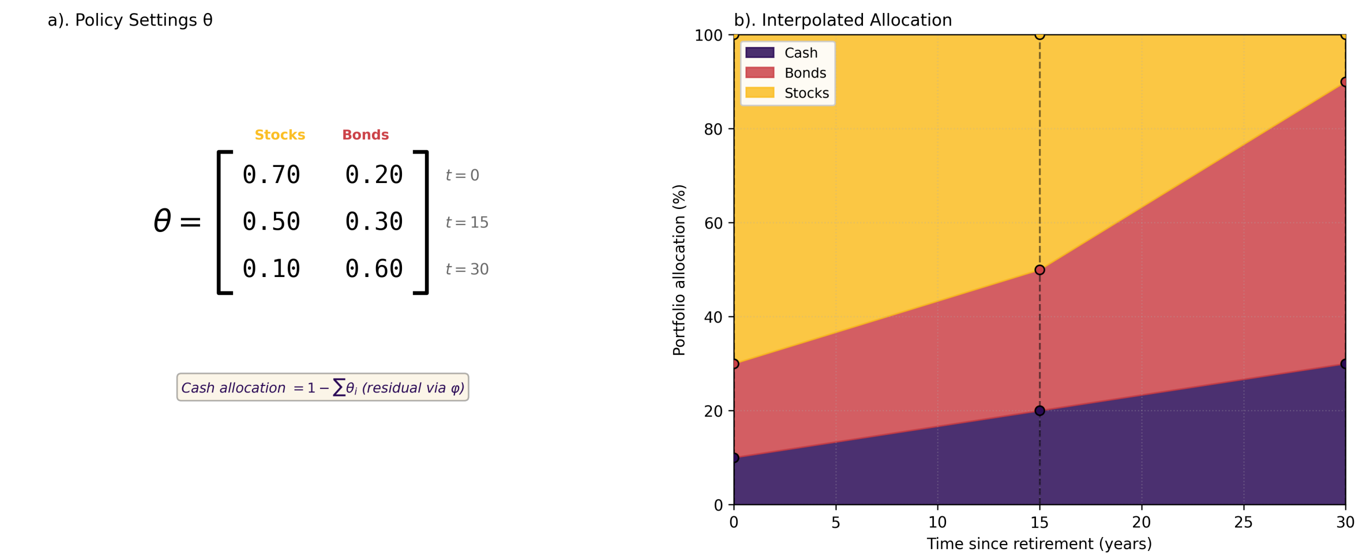

Figure 3: Linear interpolation for a time-based allocation policy with three assets and three time nodes.

This approach also allows us to constuct a wealth-responsive glidepath, which (rather than being a function of time) is a function of the investor's current wealth level. In this case, the policy function can be represented as a function of wealth:

$$\mathcal{A}(\theta, w_t) = \boldsymbol{a}_t \tag{23}$$

And, similar to above, θ is a matrix of parameters that specify the allocations at discrete wealth levels or nodes ($\{w_0, w_1, \ldots, w_M\}$), with interpolation used to determine the allocation for wealth levels that fall between the defined grid points.

The final form of the policy function that we will consider is one that allows for dependence on both time and wealth, which we can represent:

$$\mathcal{A}(\theta, t, w_t) = \boldsymbol{a}_t \tag{24}$$

This directly corresponds to the control-matrix approach of Mäkinen and Toivanen (2024), where the policy settings θ are now a three-dimensional tensor of dimension $M_t \times M_w \times (k-1)$, defined on a discrete grid of time nodes $\{t_0, t_1, \ldots, t_{M_t}\}$ and wealth nodes $\{w_0, w_1, \ldots, w_{M_w}\}$. Each element $\theta_{i,j}$ specifies the risky asset allocations when time is at node t_i and wealth is at node w_j.

For any time t and wealth w such that $t_i \le t < t_{i+1}$ and $w_j \le w < w_{j+1}$, the interpolated allocation is given by bilinear interpolation:

$$\mathbf{a}_t = \mathcal{A}(\theta, t, w) = \varphi\left((1-\alpha_t)(1-\alpha_w)\theta_{i,j} + (1-\alpha_t)\alpha_w\theta_{i,j+1} + \alpha_t(1-\alpha_w)\theta_{i+1,j} + \alpha_t\alpha_w\theta_{i+1,j+1}\right) \tag{25}$$

where the interpolation weights are computed as:

$$\alpha_t = \frac{t - t_i}{t_{i+1} - t_i}, \quad \alpha_w = \frac{w - w_j}{w_{j+1} - w_j} \tag{26}$$

This bilinear interpolation ensures that the policy varies smoothly across both time and wealth dimensions while requiring only $M_t \times M_w \times (k-1)$ parameters. The transformation φ is then applied to the interpolated risky allocations to obtain the full allocation vector including the safe asset.

To illustrate this more concretely, consider Figure 4, which shows a simple control matrix for a two-asset portfolio. The matrix has six time steps (0 to 5) and five wealth levels (0 to 4), giving a 6 by 5 grid. Each cell corresponds to a risky-asset allocation (with the remainder allocated to the safe asset). For instance, if an investor is at time step 2 with wealth level 3, the policy returns the allocation in that cell. For time or wealth values between grid points, we use bilinear interpolation to determine the allocation, which allows for a smooth transition between different allocation strategies across the state space. For example, if an investor is at time step 2.5 with wealth level 3.5, we would interpolate between the four surrounding cells to determine the appropriate allocation.

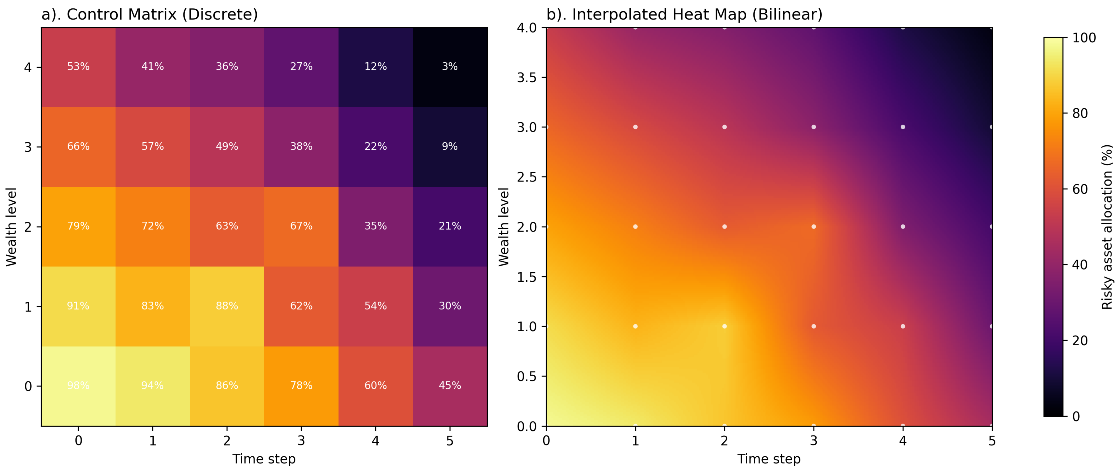

Figure 4: Illustrative example of a discrete control matrix (a) and bilinearly interpolated allocation surface (b).

The key insight is that the dimensions of θ correspond to the different state variables that the policy can depend on. Here we have extended it to two dimensions, time and wealth, which allows for policies that can adjust allocations based on both the investor's current age (or time since retirement) and their current wealth level. It is also possible to extend this approach to allow for dependence on additional state variables, such as health status or macroeconomic indicators, by adding additional dimensions to the control matrix. These can be both discrete and continuous, and the use of interpolation allows us to have a parsimonious parameterisation of the policy function while still allowing for a high degree of flexibility in the shape of the asset allocation strategy across different state variables. Compared to a neural network approach (as in Duarte et al. (2021)), the control-matrix approach allows for a more interpretable policy representation, as the parameters directly correspond to allocations at specific points in the state space, and the interpolation provides a clear mechanism for how the policy varies across different states. However, it is probably less flexible and will have a higher number of parameters than a neural network at higher dimensions, which is a trade-off that we are willing to make for the sake of interpretability in this analysis.

4.3.1 Optimisation of Asset Allocation Policies

The optimisation of θ is the core of our analysis, as it allows us to find the asset allocation strategy that maximises our chosen risk measure of the resulting wealth trajectories. Mäkinen and Toivanen (2024) use an adjoint method to compute the gradient of the risk measure with respect to the policy parameters, which allows for efficient optimisation even in high-dimensional parameter spaces. This is a powerful and efficient approach, but it is also quite complex to implement and limits the potential range of cost functions that can be used as the risk measure to those that only depend on the terminal wealth distribution. For this reason, we have implemented a more straightforward approach that relies on automatic differentiation using the PyTorch Python library (Paszke et al., 2019). This allows us to compute the gradient of the risk measure with respect to the policy parameters without having to derive and implement the adjoint equations, and it also allows us to use a wider range of risk measures that can depend on the entire wealth trajectory rather than just the terminal wealth distribution. The trade-off is that this approach is likely to be less computationally efficient than the adjoint method, particularly as the dimensionality of the policy parameter space increases, but we believe that the increased flexibility in terms of risk measures and ease of implementation makes it a worthwhile trade-off for our analysis.

The construction of our simulation framework has been carefully designed to be fully differentiable (using PyTorch's automatic differentiation capabilities) so that we can compute the gradient of our chosen risk measure with respect to the policy parameters θ and use this gradient to perform optimisation using a gradient-based algorithm such as stochastic gradient descent or Adam (Kingma & Ba, 2014). This allows us to efficiently search for the optimal policy parameters that maximise our chosen risk measure, even in high-dimensional parameter spaces.

To constrain the policy parameters θ to satisfy the allocation constraints from Equation 17 (i.e., that the allocations must be non-negative and sum to at most one), we project the parameters onto the feasible set after each gradient update using the simplex projection algorithm of Duchi et al. (2008). This efficient $\mathcal{O}(n \log n)$ algorithm finds the Euclidean projection onto the constraint set, ensuring that policy parameters remain feasible while staying as close as possible to the unconstrained gradient update. The projection preserves the optimality conditions for constrained optimisation, guaranteeing convergence to a local optimum on the constraint boundary under standard regularity conditions. A detailed description of the algorithm is provided in Appendix A.

5 Methods and Data for Simulating and Bootstrapping Asset Returns

Central to our analysis is the ability to simulate a large number of realistic asset return trajectories. As previously noted, there are two broad approaches to simulating returns: synthetic return generation using a parametric model, and bootstrapping from historical returns. The advantage of synthetic return generation is that it allows us to explore a wider range of possible return dynamics and also to create a larger number of return trajectories than would be possible with historical data alone. The disadvantage is that it relies on strong assumptions about the return-generating process, which may not or may not be an accurate representation of reality. For example, the easiests and therefore most commonly used model is the geometric Brownian motion (GBM) model, which assumes that returns are normally distributed and independent over time. In reality, asset returns exhibit features such as volatility clustering, fat tails, and serial correlation, which are not captured by the time-independent GBM model. Time series models such as GARCH can capture some of these features, but they also introduce additional complexity and make their own set of assumptions that may not hold in practice. Bootstrapping from historical returns, on the other hand, allows us to capture the empirical distribution of returns without making strong parametric assumptions. However, it is limited by the length and quality of the historical data. Underpinning both approaches is the assumption that the future will be similar to the past; in the case of synthetic return generation, this is because the model parameters are typically estimated, informed, or evaluated against historical data; in the case of bootstrapping from directly resampling from historical returns. This assumption may not hold if there are structural changes in the economy or financial markets that lead to different return dynamics in the future. This limitation is inherent to this type of research, and it forms the underlying uncertainty in any conclusions we draw from our simulations.

We will aim to generate a large number of return trajectories using both approaches, and we will evaluate the performance of different asset allocation strategies under both sets of return simulations. We will use three asset classes in our analysis: short-term bills (cash), fixed income (bonds), and equities. These asset classes are chosen because they represent the main building blocks of a typical retirement portfolio.

5.1 Geometric Brownian Motion (GBM) Model for Synthetic Return Generation

Consilium has historically used a simple geometric Brownian motion (GBM) model to generate synthetic returns for its analysis. Because the company already has established expectations of long-run returns and volatilities for different asset classes in its internal models, the GBM model is a convenient way to generate return trajectories that are consistent with these expectations.

Consilium's GBM model does not simulate the returns of the underlying assets directly, but rather simulates the returns of the portfolio as a whole. The key inputs to the GBM model are the expected return and covariance matrix of the portfolios. Expected returns of the portfolio are derived from the expected returns obtained from the factor-analysis of the underlying assets. Covariances are obtained from back-filled historical monthly portfolio returns going back to 1995.

Our approach differs from Consilium's typical implementation of the GBM model in that we will simulate the returns of the underlying asset classes (bills, bonds, equities) rather than simulating the returns of the portfolio directly. This allows us to generate return trajectories for each asset class, which can then be combined according to different asset allocation policies to produce portfolio return trajectories. This is important for our analysis because we want to continuously vary the asset allocation across the three asset classes, which is not possible if we only simulate portfolio returns directly. By simulating the underlying asset class returns, we can evaluate how different asset allocation strategies perform under a wide range of return dynamics for each asset class. The disadvantage of this approach is that the underlying composition of the asset classes cannot vary across different risk levels. For instance, Consilium's internal models vary the allocation to domestic and international equities and bonds across the different risk profiles, while our approach will use a single aggregate return series for equities and bonds. We will base our aggregate return series for each asset class on Consilium's Classic 70/30 model portfolio allocation, which represents a fairly even split between the allocations in Consilium's more and less risky model portfolios, which will mean that the asset class return dynamics we simulate will be broadly, but not perfectly, consistent with the underlying return-generating process used in Consilium's internal models.

We derive the expected returns and covariance matrix following the same approach as Consilium, using the factor-analysis of the underlying assets to obtain expected returns for each asset class, and using back-filled historical monthly returns to estimate the covariance matrix. We will then use these inputs to simulate return trajectories for each asset class using the GBM model. Consilium has traditionally treated CPI inflation as an exogenous variable that is not part of the return-generating process, but we will include it as an additional asset class in our simulations, which allows us to capture the correlation between inflation and asset returns and to evaluate how different asset allocation strategies perform under different inflation scenarios.

The expected gross returns and covariance matrix for the three asset classes and CPI inflation are shown in Table 4. These parameters are based on Consilium's internal models and are consistent with the return dynamics that we will simulate using the GBM model.

Table 4: GBM input parameters for gross returns and covariance matrix.

	Equities	Bonds	Cash	CPI
Gross return	9.31%	4.26%	1.85%	2.50%

	Cash	Bonds	Equities	CPI
Cash	0.00074818	0.00107890	-0.00011636	0.00001394
Bonds	0.00107890	0.00327030	-0.00015073	-0.00025973
Equities	-0.00011636	-0.00015073	0.01928772	-0.00051254
CPI	0.00001394	-0.00025973	-0.00051254	0.00025696

5.2 Block Bootstrap Methodology

Our block bootstrap implementation follows Anarkulova et al. (2025), adapted for a different dataset and asset allocation structure. The key advantage of the block bootstrap is that by resampling contiguous blocks of historical returns rather than individual observations, we preserve the short-run time-series dependencies and cross-sectional correlations present in asset returns while generating a large and diverse set of synthetic paths for Monte Carlo simulation.

5.2.1 Data Sources and Asset Class Construction

We construct return series for four variables—Equities, Bonds, Bills, and CPI inflation—using annual data from the Jordà-Schularick-Taylor (JST) Macrohistory Database (Jordà et al., 2017), which provides total returns and macroeconomic variables for 18 advanced economies from 1870 to the present. This long-run panel data allows us to capture return dynamics across multiple business cycles, market regimes, wars, and financial crises.

At each bootstrap iteration, we randomly designate one country as the "domestic" or "home" country, representing the New Zealand investor's perspective. Each asset class is then constructed as a weighted combination of domestic and international returns, reflecting realistic portfolio allocations for a small open economy. Specifically, each asset class k is characterized by two parameters: home bias $\omega^{(k)}$ specifying the allocation to the domestic market, and the hedging ratio $\eta^{(k)}$ specifying what fraction of the foreign exposure is currency-hedged. The asset class return combines domestic returns, currency-hedged international returns, and unhedged international returns according to these weights. We calibrate these parameters based on Consilium's typical 70/30 model portfolio allocations as shown in Table 5.

Asset Class	**Home Bias** (ω)	**Hedging Ratio** (η)
Equities	0.20	0.50
Bonds	0.33	1.00
Bills	1.00	1.00
CPI	1.00	0.00

Table 5: Asset class construction parameters reflecting realistic portfolio composition for New Zealand retail investors.

International equity returns are constructed as market-capitalization-weighted aggregates across eligible foreign countries, based on weights from the World Bank's World Development Indicators (World Federation of Exchanges, 2024) supplemented with historical estimates from Global Financial Data (sourced from Taylor, 2023). International bond returns use weights based on GDP and government debt-to-GDP ratios from the JST database. Currency conversions are performed via USD as the intermediary currency, and hedged returns eliminate currency risk while unhedged returns include both asset and currency components. Detailed descriptions of the eligibility criteria, weight construction, currency conversion methodology, and asset class formulas are provided in Appendices C–F.

5.2.2 Sampling Procedure

To generate bootstrapped return paths, we resample contiguous blocks of years from the historical data, where each block corresponds to a continuous period within a single country's history. The JST database has gaps due to wars, data quality issues, and currency crises, so we first identify all continuous periods and prevent sampling across discontinuities.

For each simulated path of length $T = 40$ years, we repeatedly draw random blocks until we accumulate the required number of observations. Each block starts at a randomly selected position within a continuous country period, and has a length drawn from a Poisson distribution with mean $\lambda = 10$ years. Blocks are truncated at period boundaries to prevent artificial transitions between countries. We pre-generate approximately 50,000 paths of 40 years each by partitioning a large batch of 2 million block draws into non-overlapping segments.

This procedure ensures that each bootstrap path preserves the empirical autocorrelation structure within blocks (capturing business cycle dynamics and mean reversion), the cross-sectional correlation among asset classes and inflation, and the time-varying composition of the global investment opportunity set. The detailed sampling algorithm, period identification procedure, and implementation notes are provided in Appendix G.

We consider two historical periods for our bootstrapping: the full sample from 1870 to the present, which captures a wide range of economic regimes and return dynamics; and a more recent sample from 1950 to the present, which may be more relevant for contemporary investors but is shorter and less diverse. By comparing results across these two samples, we can assess the sensitivity of our conclusions to different historical contexts and the potential impact of structural changes in the economy and financial markets over time.

5.3 Tax and Fees

The treatment of taxes and fees is an important consideration in retirement planning, as they can have a significant impact on the net returns that investors receive and therefore on the optimal asset allocation strategy. Retirees do not typically access securities directly, but rather invest through a retirement savings vehicle. Even after the age of 65, when they can withdraw from their KiwiSaver, most retirees will continue to invest through their KiwiSaver or some other type of managed fund, which means that they will be subject to the fees charged by these funds. The tax treatment of these assets is an incredibly complex topic, and it is beyond the scope of this report to provide a comprehensive analysis of it.

For example, dividends paid from assets held in foreign markets are subject to both foreign and domestic taxes, and the interaction between these taxes can be complex. In some cases, investors may be able to claim a foreign tax credit for the taxes paid on foreign dividends, which can reduce their overall tax burden. Further complicating this are the various tax rules around FIF and PIE investment vehicles that are commonly used by New Zealand investors to access international markets. Making a full analysis of the tax implications of different asset allocation strategies would require a detailed understanding of both the domestic and international tax rules, data about distributions and tax rates for different asset classes, and a careful modelling of how these taxes would apply to the returns generated by different asset allocation strategies. This is not something that either of our models or datasets have the capability to do. For this reason, we have chosen to use a simplified approach to taxes and fees in our analysis, which is to apply a flat fee and tax rate to the returns generated by our simulations, estimated from Consilium's internal models. While, in reality, the effective tax and fee rates that investors face will vary across asset classes and over time, this simplified approach allows us to capture the general impact of taxes and fees on net returns without having to model the complex interactions between different tax rules and asset class characteristics.

We use a flat fee rate to approximate the fees and taxes that investors would face on their returns. This fee rate is estimated from Consilium's Classic 70/30 model portfolio, and includes estimates for fund management fees, taxes on dividends and interest, foreign tax credits, domestic costs, rebalancing costs, and other relevant fees tied to brokerage. The fee rate is applied to the returns generated by our simulations to obtain net returns, which are then used in our optimisation of asset allocation strategies. This approach is crude and will understate tax in good years and overstate it in bad years, but it allows us to capture the general impact of taxes and fees on net returns without having to model the complex interactions between different tax rules and asset class characteristics. These rates are shown in Table 6.

Asset Class	**Estimated Tax and Fee Rate**
Equities	1.44% per year
Bonds	1.41% per year
Bills	0.61% per year

Table 6: Estimated flat tax and fee rates applied to simulated returns for each asset class.

6 Optimal Retirement Asset Allocation Strategies for New Zealand Retirees

Having established the simulation and optimisation framework, we now apply it to evaluate candidate lifetime allocation policies in the New Zealand retirement context.

6.1 Asset Allocation in Zero Dimensions: Fixed Allocation Strategies

6.2 Asset Allocation in One Dimension: Time-Based Glide-Paths

6.3 Asset Allocation in One Dimension: Wealth-Based Glide-Paths

6.4 Asset Allocation in Two Dimensions: Time and Wealth-Based Control Matrix

7 Discussion and Implications

8 Conclusion

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A Simplex Projection Algorithm for Constrained Optimisation

To enforce the allocation constraints in Equation 17 during gradient-based optimisation, we use the simplex projection algorithm of Duchi et al. (2008). This algorithm solves the Euclidean projection problem:

$$\text{proj}_{\Theta}(\mathbf{x}) = \arg\min_{\mathbf{y}\in\Theta} \|\mathbf{y} - \mathbf{x}\|^2 \tag{27}$$

where $\Theta = \{\mathbf{y} \in \mathbb{R}^{k-1} : y_j \geq 0, \sum_j y_j \leq 1\}$ is the feasible set for risky asset allocations (with the safe asset allocation determined residually).

A.1 Algorithm

The algorithm proceeds as follows:

1. Sort the elements of $\mathbf{x}$ in descending order to obtain $x_{(1)} \geq x_{(2)} \geq \cdots \geq x_{(k-1)}$.
2. Find the threshold λ such that the projection is given by:

$$y_j = \max(0, x_j - \lambda) \tag{28}$$

3. The threshold λ is chosen such that $\sum_j y_j = 1 - \epsilon$ for some small $\epsilon > 0$. We use $\epsilon = 10^{-8}$ to ensure the sum is strictly less than one, which ensures the safe asset receives a non-negative residual allocation: $1 - \sum_j y_j \geq 0$.
4. To find λ, identify the largest index k^* such that:

$$x_{(k^*)} - \frac{1}{k^*}\left(\sum_{j=1}^{k^*} x_{(j)} - (1-\epsilon)\right) > 0 \tag{29}$$

5. Then set:

$$\lambda = \frac{1}{k^*}\left(\sum_{j=1}^{k^*} x_{(j)} - (1-\epsilon)\right) \tag{30}$$

A.2 Properties and Computational Complexity

This projection has several desirable properties for our optimisation problem:

- **Feasibility:** After each gradient step, the policy parameters remain feasible (non-negative and summing to at most one) while staying as close as possible to the unconstrained gradient update in Euclidean distance.
- **Optimality preservation:** The projection preserves the optimality conditions for constrained optimisation. If the unconstrained optimum lies outside the feasible set, the projected gradient descent algorithm is guaranteed to converge to a local minimum on the constraint boundary under standard regularity conditions, provided the step size is appropriately chosen.
- **Computational efficiency:** The algorithm runs in $\mathcal{O}(n \log n)$ time due to the sorting step, where n is the number of risky assets. For our typical applications with 2–3 risky asset classes, this overhead is negligible.

This approach allows us to perform unconstrained gradient-based optimisation while ensuring that all iterates remain within the feasible allocation space, which is critical for the economic interpretability and practical implementation of the resulting policies.

B Replication and Discussion of Control Matrix Optimisation as Proposed by Mäkinen and Toivanen (2024)

Mäkinen and Toivanen (2024) formulate their Monte Carlo optimisation in a way that is similar to the framework described in the main text. As in our setup, they use a control matrix that allows asset allocation to respond jointly to age and wealth. Their objective function is based on terminal wealth moments, specifically variance or semi-variance. They then compute gradients of this objective with respect to control-matrix parameters using adjoint methods, which enables efficient gradient-based optimisation (in their case, a quasi-Newton method with BFGS updates).

Because this method is mathematically elegant and computationally efficient, we have attempted to replicate it here. The code for this replication can be found in the accompanying repository, and the results are shown in Figure 5 for the mean-variance and mean-semi-variance cases.

In line with the setup described by Mäkinen and Toivanen (2024), the replication uses a Monte Carlo horizon of $T = 20$ years with time step $\Delta t = 0.25$, giving $N = 80$ time steps. The policy is represented on a time-wealth control grid with $M = 31$ wealth nodes, upper wealth bound $W_{\max} = 30$, and bilinear interpolation between nodes. In the notebook, wealth is clipped to $[0, W_{\max}]$ before interpolation. Optimisation is performed with L-BFGS-B (a quasi-Newton method) with box constraints.

For the one-risky-asset, no-leverage case used in Figure 5, the full parameterisation is summarised in Table 7.

Table 7: Parameter setup for the one-risky-asset mean-variance replication.

Parameter	Value
Risk-free rate (r)	0.03
Risky-asset drift (μ_1)	0.0795
Risky-asset volatility (σ_1)	0.15
Contribution rate (π)	0.1
Initial wealth (w_0)	1
Minimum risky allocation ($p_{\min}$)	0
Maximum risky allocation ($p_{\max}$)	1
Number of simulation paths (K)	10^5
Risk-aversion weight (λ)	0.1

The replication results are broadly consistent with those reported by Mäkinen and Toivanen (2024). However, they also highlight important limitations. First, as an objective for a retirement investor, terminal-wealth variance is not especially intuitive. In our replication, the terminal-wealth distribution becomes bimodal because the optimiser shifts to a highly conservative policy at high wealth levels, truncating the upper tail that would normally appear under a log-normal-like process. Although this behaviour is mathematically consistent with variance minimisation, it is not obviously aligned with investor welfare. A similar pattern appears under semi-variance, albeit at higher wealth levels. Because semi-variance is computed relative to the mean, higher-return strategies can increase the penalty by raising the mean itself, again encouraging upper-tail truncation. This reinforces the need for risk measures that are both interpretable and behaviourally relevant.

Additionally, the control matrix has limited effective resolution in early periods: many simulated paths map to only a few low-wealth nodes because initial wealth is far below the fixed upper wealth bound. This reduces the optimiser's ability to distinguish trajectories precisely when policy differences likely matter most due to compounding. It also creates many redundant parameters at wealth levels that are never reached (upper-left region in Figure 5). For this reason, we propose calibrating the control-matrix upper bound to the simulated wealth distribution so that resolution is concentrated where paths actually lie.

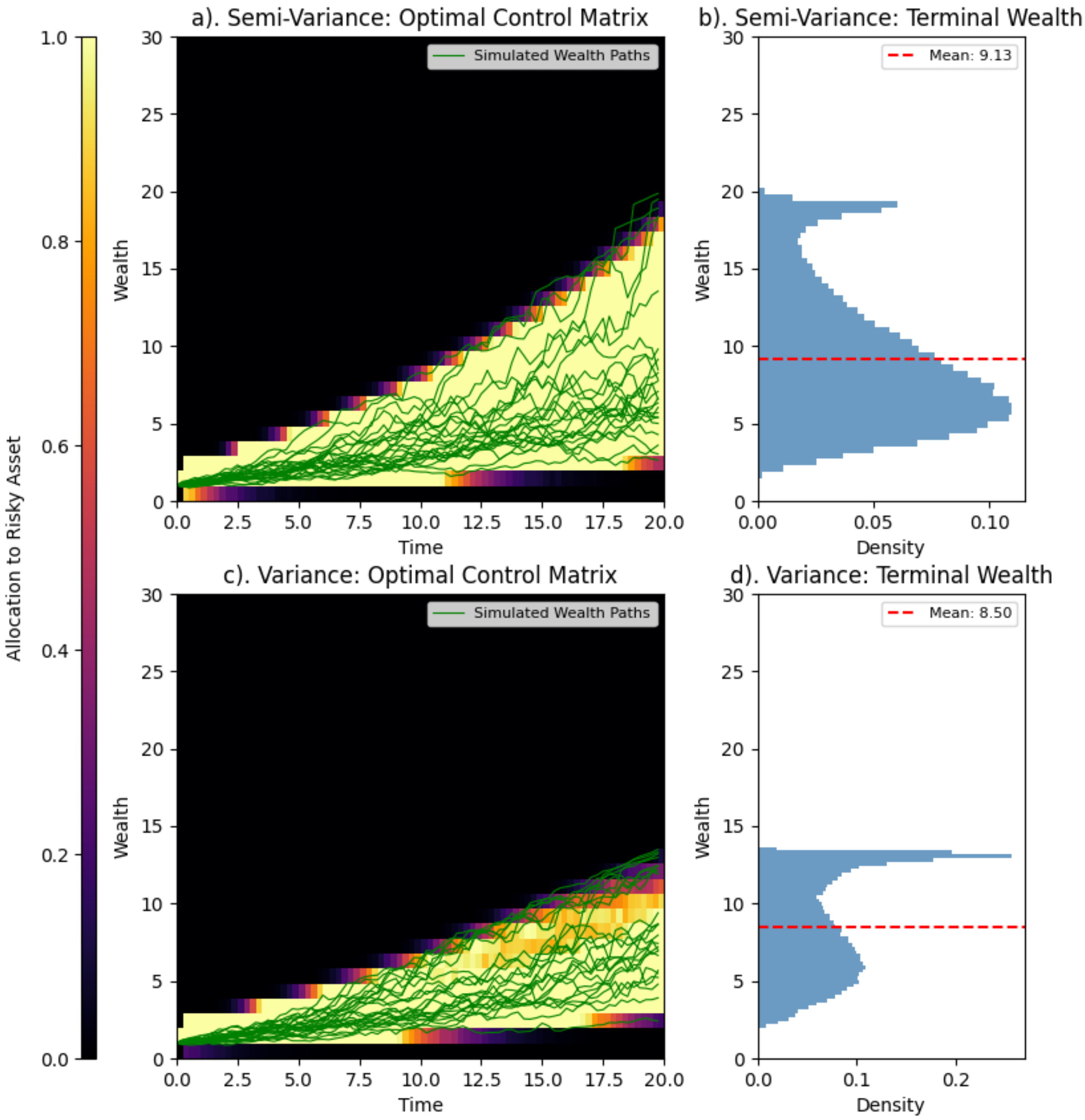

Figure 5: Replication of the control matrix optimization method proposed by Mäkinen and Toivanen (2024) for one risky asset and no leverage. The left column shows the optimal control matrix for the mean-semi-variance case (top) and the mean-variance case (bottom). The right column shows the corresponding histograms of terminal wealth for each risk measure respectively.

C Block-Bootstrapping: Investable Universes

To implement the block bootstrap methodology for New Zealand retirees, we must first construct appropriate investable universes from the JST database. A critical insight from Anarkulova et al. (2025) is that the bootstrap procedure should reflect the investment opportunity set available to the domestic investor, which differs from simply using aggregate global indices. Specifically, at each bootstrap iteration, we designate one country as the "home" or "domestic" country, representing the investor's perspective, and construct return series for three asset classes relative to that country's viewpoint: domestic short-term government bills (cash), domestic intermediate-term government bonds, and international equities.

This approach requires us to define, for each year in the historical data, which countries can serve as domestic markets and which can serve as sources of foreign returns. These definitions are not uniform across time, as data availability and quality vary significantly across countries and periods. The following sections describe how we construct these investable universes and define the corresponding asset classes.

C.1 Data Quality Requirements and Eligibility Criteria

For a country to be eligible as a domestic market in a given year, we require complete and reliable data for all key return series needed to construct a domestic investor's portfolio. Specifically, a country can serve as "domestic" only if the following conditions are met:

1. **Complete asset return data:** The country must have non-missing values for equity total returns (`eq_tr`), bond total returns (`bond_tr`), and short-term bill rates (`bill_rate`) in that year.

2. **Inflation data:** The country must have a valid inflation rate (computed from CPI) to allow for real return calculations.

3. **Exchange rate data:** The country must have a valid exchange rate against the US dollar (`xrusd`) and its percentage change (`xrusd_return`), which is necessary for converting foreign returns to the domestic investor's currency (and vice versa).

4. **Market size constraint:** The country's equity market capitalization must not exceed 20% of the total market capitalization across all JST countries in that year. This threshold prevents the United States and United Kingdom being selected as domestic markets, as their inclusion would significantly reduce the diversity of bootstrap samples and make the simulated paths less representative of smaller developed economies like New Zealand.

For a country to be eligible as a source of foreign returns (either equity or bonds), we require non-missing asset return data (equity total returns `eq_tr` for the foreign equity universe, or bond total returns `bond_tr` for the foreign bond universe), valid exchange rate data (`xrusd_return`), and three-year rolling cumulative exchange rate depreciation not exceeding 50%.

The 50% cumulative depreciation threshold over a three-year rolling window serves to exclude protracted currency-crisis episodes that, while historically occurred, create irregularities in the return distribution that are not representative of normal market dynamics. Currency crises often unfold over multiple years rather than single-year shocks, so a multi-year filter is more effective at identifying and excluding crisis periods. It is unlikely that a typical investor would maintain exposure to a foreign market experiencing such extreme sustained currency depreciation, so excluding these episodes makes the simulated return paths more realistic for retirement planning purposes. However, we choose to retain these episodes in the domestic return series when they occur in the home country, as they are part of the historical experience that a domestic investor would have faced and cannot be avoided through diversification.

Figure 6 displays the time periods during which each JST country satisfies the eligibility criteria for domestic, foreign equity, and foreign bond status. Panel (a) shows that most countries have continuous domestic eligibility from the mid-20th century onward. There are some gaps in eligibility for certain countries, the USA, Japan and the UK get excluded at times due to the market-cap constraint, and some countries have shorter data histories that limit their eligibility in earlier years. Germany loses foreign equity eligibility during the hyperinflation episode in the early 1920s, and many other gaps are present due to wars, regime changes, or data quality issues.

C.2 Construction of Investable Universes

Given the eligibility criteria defined above, we construct two investable "universes" that define the valid constituents for the international equity and bond return series in the block bootstrap simulations:

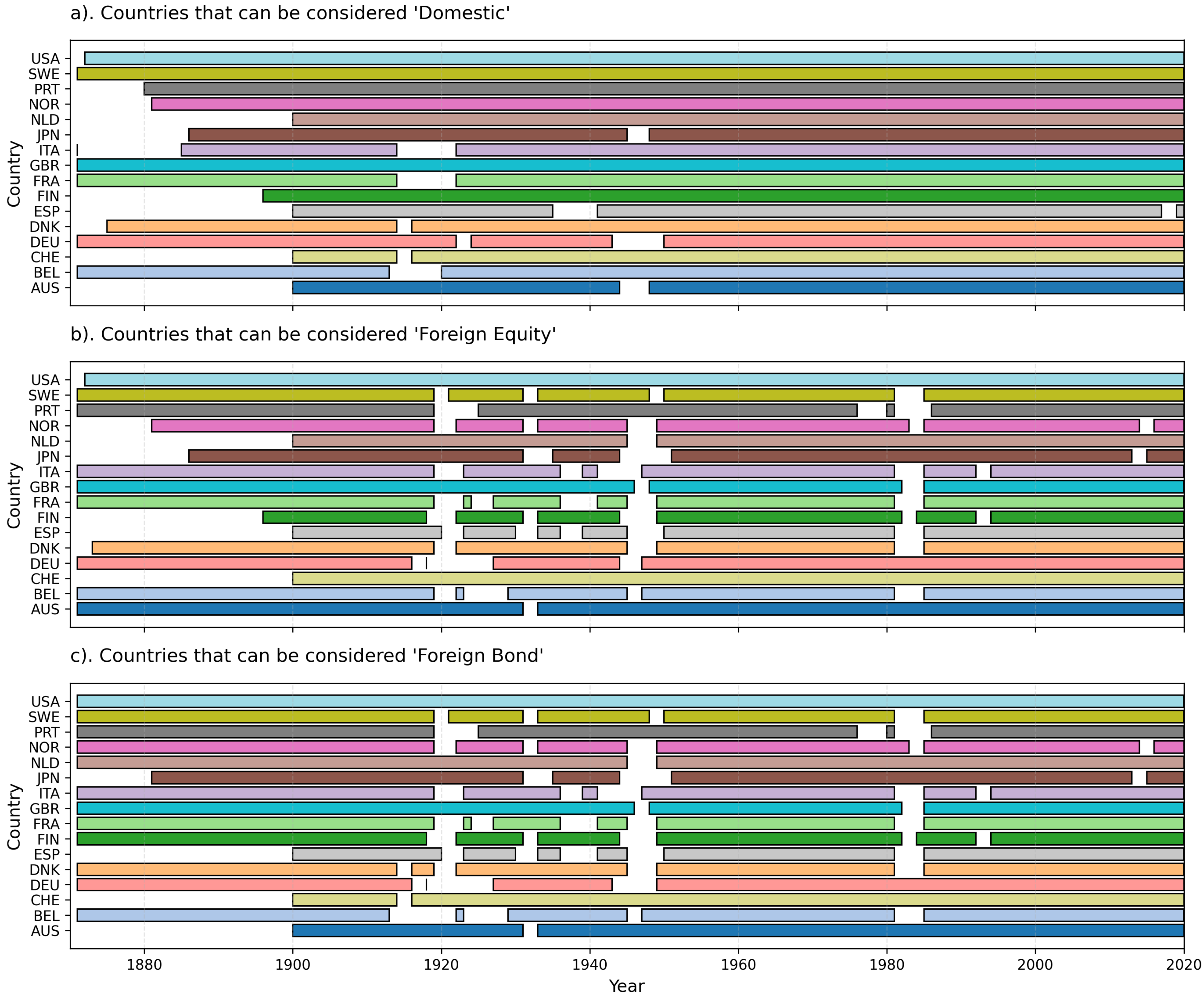

Figure 6: Eligibility of JST countries as domestic markets (a), foreign equity sources (b), and foreign bond sources (c) over time.

1. **Equities universe:** For each year t, this universe consists of all ordered pairs (i, j) where country i is eligible as domestic and country j is eligible as foreign equity in year t, subject to the constraint that $i \neq j$ (i.e., a country cannot be both the domestic and foreign market in the same bootstrap draw). This universe defines the feasible combinations for constructing the international equity return series from the perspective of domestic country i.

2. **Bonds universe:** Analogously, for each year t, this universe consists of all ordered pairs (i, j) where country i is eligible as domestic and country j is eligible as foreign bonds in year t, with $i \neq j$. This universe defines the feasible combinations for constructing the international bond return series.

At each iteration of the block bootstrap, we randomly select a domestic country i from those eligible in the starting year of the block, then construct foreign equity and bond return series as weighted averages of the returns from all countries j in the corresponding universe for that domestic country and year. This approach ensures that the simulated return paths reflect the diversification opportunities and correlation structure available to the domestic investor, while also preserving the time-varying composition of the global investment opportunity set.

D Block-Bootstrapping: Equity Market Weights

Following Anarkulova et al. (2025), our implementation of the block bootstrap method for simulating returns relies on constructing a market-capitalisation-weighted aggregate return series for international securities. To do this, we need to construct a timeseries of market capitalization weights for the countries in the JST database. This is necessary because the JST database does not include equity market capitalization weights.

We define equity market capitalization weights as the ratio of a country's total market capitalization to the total market capitalization of the country cohort. To construct the baseline capitalization weights, we combine data from the World Bank's World Development Indicators (WDI) database and historical estimates from Global Financial Data (GFD). The World Bank provides market capitalization of listed domestic companies (in inflation-adjusted US Dollars) from the World Federation of Exchanges database (World Federation of Exchanges, 2024). This dataset covers 1975–2024 with good coverage for most developed economies. The data is publicly available under a CC BY-4.0 license and represents the most authoritative source of the two used in our analysis for market capitalization figures. However, the WDI dataset does not cover the historical period of interest for our analysis (1870–1975), so we supplement it with estimates from GFD, which provides long-run historical estimates of market capitalization weights for major economies.

The GFD database provides historical estimates of a number of macro-financial variables, including market capitalization weights for major economies. This is the dataset leveraged by Anarkulova et al. (2025) to construct their international equity return series. However, we do not have direct access to the GFD database, so we rely on a published chart from Finaeon's website (sourced from Taylor, 2023) that displays market capitalization weights a number of countries from 1782 to 2022. We extract approximate values from this chart using an image processing method that counts pixels of different colors corresponding to each country. This method is subject to measurement error and should be interpreted with caution, but it provides a rough estimate of the historical market capitalization weights for the countries in the JST database.

The World Bank data processing is straightforward and follows standard data cleaning procedures:

1. Load the raw CSV file containing market capitalization values in current US$ for all countries
2. Filter to include only the JST countries
3. For each year, calculate the total market capitalization across all JST countries
4. Compute each country's weight as its market cap divided by the total
5. Apply a 2% minimum weight floor to ensure diversification
6. Re-normalize weights to sum to 1.0 after applying the floor

The extraction of historical market capitalization weights from the GFD database presents significant methodological challenges. Without direct access to the underlying GFD database, we must rely on a published chart from Taylor (2023) that displays market cap percentages as a stacked area graph. The extraction procedure involves the following steps:

1. Crop the chart image to isolate the plotting area
2. Map pixel x-coordinates to years using the known chart range (1782–2022)
3. For each year, identify pixels corresponding to each country color
4. Calculate the vertical percentage occupied by each color
5. Apply a centered 3-year rolling average to smooth pixel-counting noise
6. Distribute the unclassified "Other" percentage evenly across all the JST countries.
7. Apply forward-fill for missing values

This introduces several sources of uncertainty and measurement error. Because we extract percentages by analyzing the pixel colors in the chart image, each country is represented by a distinct color, and we count pixels of each color at each year's x-coordinate position. This approach is fundamentally limited by the chart resolution and image quality, interpolation between discrete x-axis positions, anti-aliasing and edge effects at color boundaries, and potential color bleeding or compression artifacts. For these reasons, the pixel-counting method produces noisy estimates. To mitigate this, we apply a 3-year centered rolling average to reduce this noise, but which may over-smooth genuine historical variation.

Furthermore, the original chart displays only five countries relevant to the JST data (USA, UK, Germany, France, and Japan), with all other countries aggregated into an "Other" category. To construct weights for all JST countries, we must distribute this unclassified mass evenly across the remaining 13 countries. This is a strong and likely unrealistic assumption, as it ignores the actual variation in market capitalizations among these countries. Finally, because we do not have access to the original GFD database, we cannot validate our extracted values against the true underlying data. The accuracy of our historical estimates is therefore unknown, but what data we have extracted roughly aligns with the World Bank data in the overlapping period (see Figure 7). The extracted GFD data shows reasonable aggregate patterns (e.g., U.S. dominance in the mid-20th century, rising Japanese share in the 1980s), but individual country estimates, particularly for the "Other" countries, should be interpreted with considerable caution. However, because the share of the "Other" category is relatively small in the later period and the goal is to capture broad historical trends rather than precise country-level weights, we consider this approach acceptable for our purposes.

The final, combined dataset of weights prioritizes the more reliable World Bank data where available and fills historical gaps with the GFD estimates. Figure 7 displays the three datasets side-by-side: GFD extracted estimates (panel a), World Bank data (panel b), and the combined final dataset (panel c). The overlap period (1988–1989) shows reasonable consistency between the two sources.

Users of this dataset should be aware of several important caveats. The market cap weights extracted from the GFD data are approximate and should not be interpreted as precise measurements. They are sufficient for characterizing broad historical trends but are unsuitable for applications requiring exact country-level weights. The even redistribution of the unclassified percentage across all JST countries is a strong simplifying assumption. In reality, countries like Australia, Canada, and Switzerland likely had substantially different market capitalizations than smaller European economies. The GFD and World Bank datasets may differ in their treatment of market entries, exits, and index composition. These methodological differences are absorbed into our combined series but are not explicitly documented. Ideally, historical market cap weights would be constructed from direct access to the GFD database or from alternative sources such as the Dimson-Marsh-Staunton Global Returns Database. However, these sources were not available for this study.

Despite these limitations, we believe the combined dataset provides a reasonable approximation of historical market capitalization weights for the JST countries and is suitable for the portfolio construction and simulation exercises undertaken in this research. The primary analyses focus on aggregate portfolio behavior rather than individual country dynamics, which reduces the impact of country-level measurement error. Nevertheless, readers should interpret historical results (particularly those prior to 1988) with appropriate caution regarding the data quality constraints described above.

All data processing code, quality checks, and intermediate outputs are available in the project repository to facilitate transparency and replication.

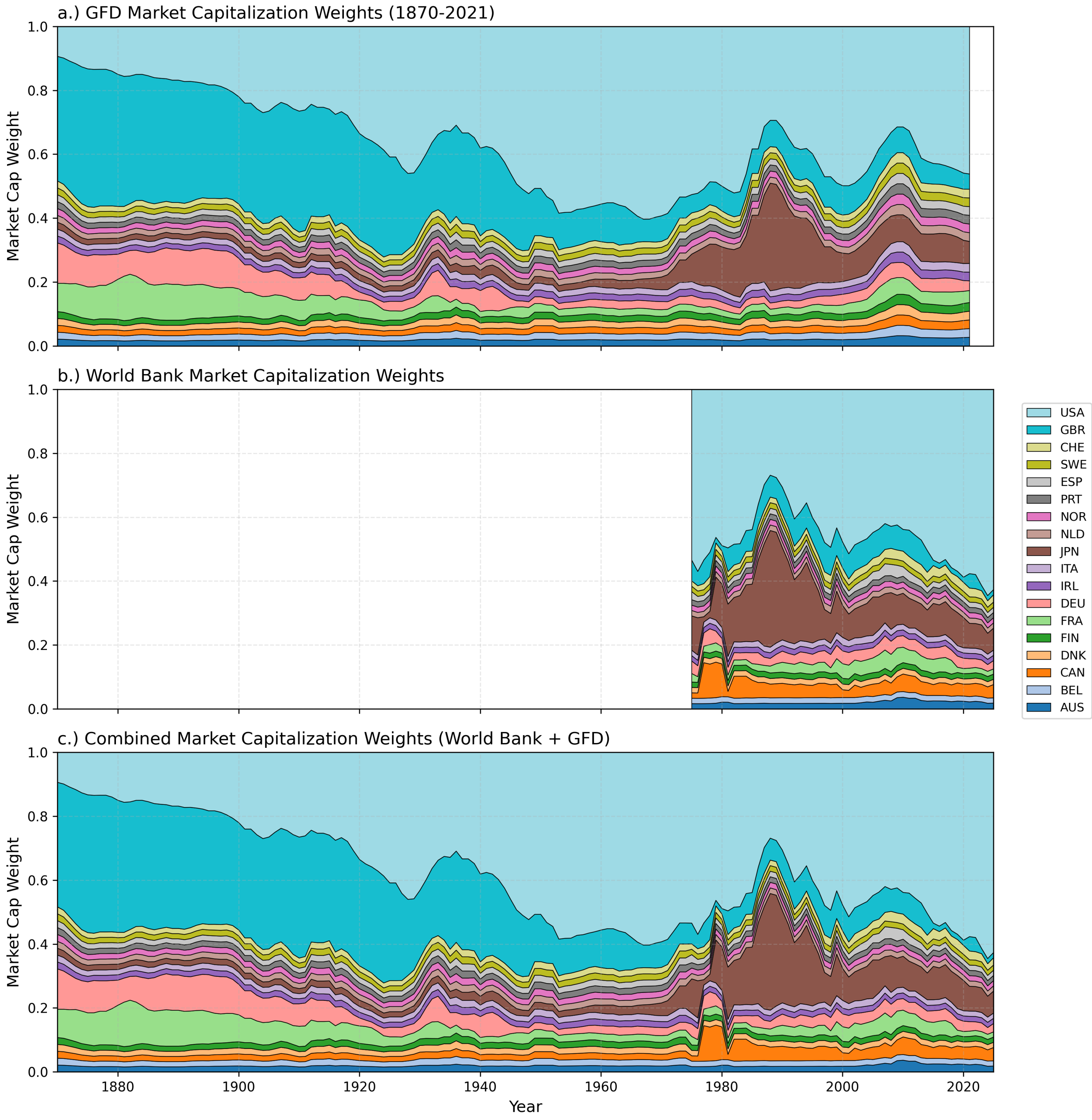

Figure 7: Market capitalization weights for JST countries from three sources. Panel (a) shows estimates extracted from the GFD historical chart (1870–2021), panel (b) shows World Bank data (1988–2024), and panel (c) shows the combined dataset using World Bank data where available and GFD estimates for earlier periods. Colors are consistent across panels to facilitate comparison.

E Block Bootstrapping: Debt Market Weights

Unlike Anarkulova et al. (2025), investors in our analysis have access to a domestic and international bond market. To construct the return series for the international bond component, we need to determine the appropriate weights for the bonds of the different countries in the JST database. Fortunately, the JST database includes a GDP and debt-to-GDP ratio for each country, which allows us to estimate the total market capitalization of each country's bond market. We then compute the weights for the international bond component as the ratio of each country's bond market capitalization (converted to USD using exchange rates from the JST database) to the total bond market capitalization across all countries. This method is a simplification, as it assumes that the bond market capitalization is directly proportional to GDP and debt levels, which may not hold in practice due to differences in financial development, government borrowing needs, and investor preferences. However, it provides a reasonable approximation for our purposes given the data constraints. The resulting bond market capitalization weights are shown in Figure 8.

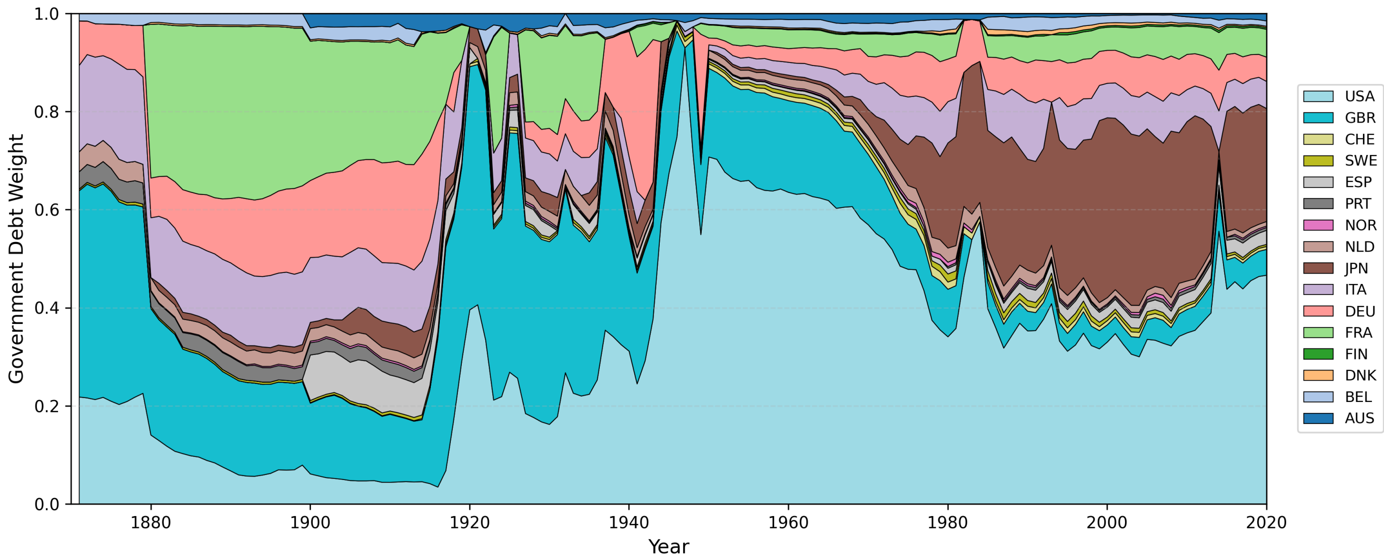

Figure 8: Estimated bond market capitalization weights for JST countries. We compute these weights using GDP and debt-to-GDP ratios from the JST database, converting to USD using the provided exchange rates.

F Block Bootstrapping: Asset Class Returns

To construct the return series for the three asset classes (Cash, Bonds, Equities) and the inflation rate for each country, we use the following variables from the JST database:

1. `bill_rate`: This variable represents the short-term government bill rate for each country. We use this as the return series for the Cash asset class.

2. `bond_tr`: This variable represents the total return on government bonds for each country (in its local currency). It is calculated by JST as $r_t = \frac{p_t + c_t}{p_{t-1}} - 1$, where p_t is the price of the bond at time t and c_t is the coupon payment received at time t. We use this as the return series for the Bonds asset class.

3. `eq_tr`: This variable represents the total return on equities for each country (in its local currency). It is calculated by JST as $r_t = \frac{p_t + d_t}{p_{t-1}} - 1$, where p_t is the price of the equity index at time t and d_t is the dividend payment received at time t. We use this as the return series for the Equities asset class.

4. `cpi`: This variable represents the annual inflation rate for each country, calculated as the percentage change in the Consumer Price Index (CPI) from one year to the next. We use this variable to compute real returns for each asset class by adjusting the nominal returns for inflation.

5. `xrusd`: This variable represents the exchange rate of the country's currency against the US dollar.

These asset class returns are not totally analogous to the asset classes used in the Consilium model portfolios. Bonds in the JST database are government bonds, whereas the Consilium portfolios include a mix of government and corporate bonds. However, given the data constraints and the focus on capturing broad return dynamics rather than precise asset class replication, we consider the JST bond returns to be a reasonable proxy for the fixed-income component of the portfolios. Similarly, the equity returns in the JST database are based on national stock market indices, which may differ from the global equity exposure in the Consilium portfolios. Additionally, Consilium's international equity exposure includes emerging markets, which are not represented in the JST database. While at some points countries in the JST database may have had emerging market characteristics, we acknowledge that our international equity return series may not fully capture the risk-return profile of a global equity portfolio with emerging market exposure. However, the JST equity returns still provide valuable insights into the historical performance of equities across different countries and time periods, which is relevant for our analysis of retirement portfolio strategies.

F.1 Foreign Return Construction and Currency Considerations

While Cash will comprise entirely of domestic bills, the Bonds and Equities asset classes will be constructed as weighted averages of the returns from eligible foreign countries, with both currency hedged and unhedged components (for equities). All returns in the JST database are reported in nominal local currency terms. To construct return series from the perspective of a domestic investor in country i, we must account for exchange rate movements when investing in foreign assets.

We introduce the following notation to clarify the currency conversions:

- $r^j_{j,t}$: Total return on an asset in country j at time t, denominated in country j's local currency (as reported in the JST database)

- $x_{j,t}$: Exchange rate return for country j's currency relative to USD at time t

- $r^{\text{USD}}_{j,t}$: Total return on an asset in country j, converted to USD

- $r^i_{j,t}$: Total return on an asset in country j, from the perspective of an investor in country i (denominated in country i's currency)

The exchange rate variable `xrusd` in the JST database represents the exchange rate from USD to local currency. We compute the exchange rate return for country j as the percentage change in this exchange rate:

$$x_{j,t} = \frac{\texttt{xrusd}_{j,t}}{\texttt{xrusd}_{j,t-1}} - 1 \tag{31}$$

This represents the return (or loss) from holding USD relative to country j's local currency. A positive $x_{j,t}$ indicates depreciation of country j's currency against the USD (this is what we use in Appendix C to exclude extreme depreciation episodes), while a negative $x_{j,t}$ indicates appreciation.

To convert a foreign asset's local currency return to USD terms, we account for both the asset return and the exchange rate movement. The total return in USD for an asset in country j is:

$$r_{j,t}^{\text{USD}} = (1 + r_{j,t}^{j})(1 + x_{j,t})^{-1} - 1 \tag{32}$$

This formulation shows that if country j's currency depreciates against the USD (positive $x_{j,t}$), the USD-denominated return will be lower than the local currency return, as the investor experiences a currency loss when converting back to USD.

From the perspective of a domestic investor in country i, we must further convert the USD return to country i's local currency. The unhedged return on a foreign asset in country j from country i's perspective is:

$$r_{j,t}^{i} = (1 + r_{j,t}^{\text{USD}})(1 + x_{i,t})^{-1} - 1 = \frac{(1 + r_{j,t}^{j})}{(1 + x_{j,t})(1 + x_{i,t})} - 1 \tag{33}$$

This expression captures the complete currency conversion chain: starting with country j's local currency, converting to USD, and then converting to country i's local currency. The investor experiences both the asset return and the net exchange rate movement between the two currencies.

For the currency-hedged return, we assume perfect hedging that eliminates all currency risk. In this case, the investor receives only the local currency return, expressed in their domestic currency terms:

$$r_{j,t}^{i,\text{hedged}} = r_{j,t}^{j} \tag{34}$$

In practice, currency-hedged returns would also include the cost of hedging (typically related to the interest rate differential between countries). For simplicity, and given data constraints, we assume this cost is negligible and that hedging is perfectly efficient in removing currency risk. This assumption is reasonable for developed markets with liquid forward currency markets, which characterizes most of the countries in the JST database during the period of our analysis.

F.2 Foreign Return Weighting and Aggregation

To construct the international component of each asset class, we aggregate returns from all eligible foreign countries using market-capitalization-based weights. For equities, we use the equity market capitalization weights described in Appendix D; for bonds, we use the government debt weights from Appendix E.

Because the eligible universe varies across time (as countries gain or lose eligibility) and across domestic perspectives (due to the constraint that $i \neq j$), we must renormalize the weights at each time step for each domestic country. For a domestic investor in country i investing in asset class k at time t, let $\mathcal{F}_{i,t}^{(k)}$ denote the set of countries eligible as foreign sources for that asset class. The renormalized weight for foreign country j is then:

$$w_{i,j,t}^{(k)} = \frac{w_{j,t}}{\sum_{\ell \in \mathcal{F}_{i,t}^{(k)}} w_{\ell,t}} \tag{35}$$

Where $w_{j,t}$ is the baseline market capitalization weight for country j at time t (before filtering for eligibility), and the denominator sums over all countries ℓ that are eligible foreign sources for domestic country i and asset class k at time t. This renormalization ensures that the weights sum to exactly 1.0 across the eligible universe.

The aggregated international return for asset class k from the perspective of a domestic investor in country i is then the weighted average:

$$r_{k,t}^{i,\text{intl}} = \sum_{j \in \mathcal{F}_{i,t}^{(k)}} w_{i,j,t}^{(k)} \cdot r_{j,t}^{i} \tag{36}$$

Where $r_{j,t}^{i}$ is the return on asset class k in country j, converted to country i's currency as described in equations 32 and 33. This aggregation preserves the empirical correlation structure among foreign markets while reflecting the time-varying composition of the global investment opportunity set.

F.3 Constructing Asset Class Returns

In our block-bootstrapping framework, each asset class k is characterized by two parameters: a Home Bias $\omega^{(k)} \in [0, 1]$ and a Hedging Ratio $\eta^{(k)} \in [0, 1]$. The home bias determines the allocation between domestic and foreign markets within that asset class, while the hedging ratio determines what fraction of the foreign exposure is currency-hedged versus unhedged. These parameters allow us to construct asset class returns that reflect realistic portfolio compositions for New Zealand investors.

The hedged international return is computed analogously to the unhedged return in equation 36, but using local currency returns without currency conversion:

$$r_{k,t}^{i,\text{hedged}} = \sum_{j \in \mathcal{F}_{i,t}^{(k)}} w_{i,j,t}^{(k)} \cdot r_{j,t}^{j} \tag{37}$$

Where $r_{j,t}^{j}$ is the asset return in country j's local currency (i.e., the same return that appears in equation 34). This represents the return from a perfectly currency-hedged foreign investment.

The final return for asset class k from the perspective of a domestic investor in country i at time t combines domestic, hedged foreign, and unhedged foreign components:

$$r_{k,t}^{i} = \omega^{(k)} \cdot r_{k,t}^{i,\text{home}} + (1 - \omega^{(k)}) \cdot \left[\eta^{(k)} \cdot r_{k,t}^{i,\text{hedged}} + (1 - \eta^{(k)}) \cdot r_{k,t}^{i,\text{intl}}\right] \tag{38}$$

Where $r_{k,t}^{i,\text{home}}$ is the return on asset class k in the domestic country i (e.g., `bill_rate` for Cash, `bond_tr` for Bonds, `eq_tr` for Equities), $r_{k,t}^{i,\text{hedged}}$ is the currency-hedged international return from equation 37, and $r_{k,t}^{i,\text{intl}}$ is the unhedged international return from equation 36 which includes currency effects. For the purposes of block bootstrapping, we treat CPI inflation as an asset class in its own right with a 100% home bias and no foreign component.

G Block Bootstrapping: Sampling Procedure

Having constructed the asset class returns for each eligible domestic country and year in Appendix F, we now describe the procedure for generating bootstrapped return paths. Our implementation follows in the spirit of Anarkulova et al. (2025) but includes some adjustments. The key idea is to sample blocks of consecutive years from the historical data, where each block corresponds to a continuous period of data for a single country. By concatenating these blocks, we create synthetic return paths that preserve the short and medium-term autocorrelation structure of returns while also capturing the cross-sectional correlation across countries and asset classes.

The first step is to identify continuous time periods within each country's history. As shown in Figure 6, many countries have gaps in their eligibility due to wars, data quality issues, or currency crises. We cannot simply sample random year ranges, as this would risk drawing blocks that span discontinuities where no data exists.

We identify continuous periods by sorting the domestic mask by country and year, then flagging years where the gap from the previous year exceeds one. A cumulative sum of these flags creates period groups within each country. For each continuous period p, we record the country identifier (iso_p), start year (y_p^{start}), end year (y_p^{end}), and period length ($L_p = y_p^{\text{end}} - y_p^{\text{start}} + 1$).

We then create a global index that maps each year of data to a single sequential index $s \in \{1, 2, \ldots, S\}$, where $S = \sum_p L_p$ is the total number of country-year observations across all periods. Each period p occupies indices from s_p^{start} to s_p^{end} in this global sequence.

To generate a single bootstrap path of length $T = 40$ years, we repeatedly draw random blocks and concatenate them until we have accumulated the required number of observations. For each block b, we:

1. Draw a random starting index $s_b \sim \text{Uniform}\{1, 2, \ldots, S\}$ from the global index
2. Draw a block length $\ell_b \sim \text{Poisson}(\lambda = 10)$ where the Poisson parameter $\lambda = 10$ sets the mean block length to 10 years
3. Identify which period p contains index s_b
4. Compute the actual block end as $s_b^{\text{end}} = \min(s_b + \ell_b - 1, s_p^{\text{end}})$ to prevent the block from extending beyond the period boundary
5. Extract the corresponding years from period p: if s_b corresponds to year y within period p, then the block spans years $y, y+1, \ldots, y + (s_b^{\text{end}} - s_b)$

This process ensures that each block respects the continuous history of a single country, preventing artificial discontinuities in the bootstrapped series.

Our implementation uses a Poisson distribution for block lengths, whereas Anarkulova et al. (2025) use a geometric distribution. Both choices target a mean block length of approximately 10 years, but they differ in their theoretical properties and practical implications. The practical impact on our analysis is minimal. Both distributions preserve the medium-run autocorrelation structure that is the primary motivation for block bootstrapping. We chose the Poisson distribution primarily for computational convenience: it is straightforward to implement and produces non-negative integer block lengths directly without requiring rejection sampling or other adjustments. This pragmatic choice does not materially affect the properties of the bootstrap procedure or the substantive conclusions of our analysis.

Blocks are concatenated sequentially until we reach T years. If a block would cause the total to exceed T, we truncate it. The result is a sequence of (i_t, y_t) pairs for $t = 1, \ldots, T$, where i_t is the domestic country and y_t is the year for observation t in the bootstrapped path.

For each (i_t, y_t) pair, we retrieve the corresponding asset class returns from the processed data created in Appendix F. This gives us a $T \times 4$ matrix of returns for the path, with columns for Equities, Bonds, Bills, and CPI (in that order).

To generate a large number of paths efficiently, we pre-generate N block draws, where $N = 2{,}000{,}000$ in our implementation. We then partition this sequence into non-overlapping segments of length T to create approximately $N/T \approx 450{,}000$ independent paths. Each path represents a plausible 40-year sequence of annual returns that preserves the short-run autocorrelation and cross-sectional correlation structure of the historical data.

The final output is a three-dimensional array of shape ($N_{\text{paths}} \times T \times 4$) where:

- Dimension 1 indexes the path (simulation)
- Dimension 2 indexes the year within the path
- Dimension 3 indexes the asset class (Equities, Bonds, Bills, CPI)

This array is saved to disk for use in the Monte Carlo simulations described in the main analysis.